

Prelude

A typical MSc thesis will be structured according to a fairly standard sequence of sections, and one example is shown here. However, it is hard and perhaps even counter-productive to generalise: the goal of this document is *not* to be prescriptive, to show you how you *must* structure your thesis; rather it is to simply to act as a guideline, to show you how you *could* structure your thesis. In particular, each chapter's indicated page-count is important but *not* absolute: the aim is simply to highlight that a clear and concise description is better than a rambling alternative that makes it hard to separate important content and facts from irrelevant trivia.

You can use this document as a L^AT_EX-based template for your own thesis by simply deleting extraneous sections and content, and adding your own text. If you do that, we recommend using BibTeX to deal with the associated bibliography. You may want to use Overleaf (an online cloud-based collaborative L^AT_EXplatform).

The words “thesis” and “dissertation” are used interchangeably here. All active hyperlinks in the PDF file are rendered in the same shade of dark blue.



DEPARTMENT OF ENGINEERING MATHEMATICS

Using machine learning to forecast physical activity participation, volunteering and activity patterns in London

Ashrey Ashrey, Sanya Kapoor, Trisha Sharma, Yash Bhooshan
Pandloskar

A dissertation submitted to the University of Bristol in accordance with the requirements of the degree
of Master of Science in the Faculty of Engineering.

Wednesday 2nd September, 2026

Supervisor: Dr. Dalila O'Grady

Declaration

This dissertation is submitted to the University of Bristol in accordance with the requirements of the degree of MSc in the Faculty of Engineering. It has not been submitted for any other degree or diploma of any examining body. Except where specifically acknowledged, it is all the work of the Author.

Ashrey Ashrey, Sanya Kapoor, Trisha Sharma, Yash Bhooshan Pandloskar, Wednesday 2nd September, 2026

Contents

1	Introduction	1
1.1	Context and Problem Statement	1
1.2	Motivation and Contribution	1
1.3	Central Challenges	2
1.4	Summary of Aims, Objectives and Achievements	2
2	Background	4
2.1	Problem Definition	4
2.2	Domain Background: Determinants of Participation	4
2.3	Forecasting Approaches Considered and Limitations of Conventional Models	5
2.4	The Borough-Forecasting Model	5
2.5	Composition–Place Decomposition & Model Interpretability	6
2.6	Alternative Approaches Explored	6
2.7	Related Work and Research Gap	7
3	Methodology	9
3.1	Overall Project Workflow	9
3.2	Dataset Description	10
3.3	Data Preprocessing	10
3.4	Exploratory Data Analysis	12
3.5	Machine Learning Framework	14
4	Results and Critical Evaluation	15
4.1	Experimental Setup	15
4.2	Model Comparison – Performance and Critical Analysis	15
4.3	Physical Activity and Volunteering Trends Over the Next Ten Years	18
4.4	Overall Synthesis	27
4.5	Limitations	27
5	Conclusion	28
5.1	Summary of Contributions	28
5.2	Evaluation Against Aims and Objectives	28
5.3	Limitations and Future Work	29
A	An Example Appendix	32

List of Figures

3.1	Overall project workflow diagram.	9
3.2	Data preprocessing workflow showing dataset harmonisation, data quality checks, feature engineering, and creation of the final modelling datasets.	10
3.3	Population-weighted demographic characteristics of respondents.	12
3.4	Population-weighted distribution of activity levels across survey years.	13
3.5	Geographical variation in population-weighted activity levels across London's boroughs.	13
3.6	Population-weighted volunteering participation across survey years.	14
4.1	LightGBM predictive performance relative to the benchmarked ordinal logistic regression.	16
4.2	LightGBM borough-level correlation (left) and calibration reliability plot (right).	16
4.3	London-wide activity bands, observed 2015–16 to 2022–23 and projected to 2025–26.	18
4.4	Ten-year scenario range for London-wide activity, 2022–2032.	19
4.5	Active travel: ten-year scenarios (widest scenario spread of any activity measured).	19
4.6	Walking: ten-year scenarios (largest-base activity measured).	20
4.7	Club membership (any sport): ten-year scenarios.	20
4.8	Ten most active boroughs and five least active boroughs, projected share of adults meeting the guideline, 2025.	21
4.9	Reasons free time for activity is limited, by age group and gender (London adults, 2022–23).	21
4.10	Activity rate by whether free time is constrained, by age group and gender.	22
4.11	Activity forecast by demographic group; bar = 3-year forecast range, dot = today's rate.	22
4.12	Which activities each group does most; bold marks the top activity for that group.	23
4.13	Age gradient within each borough, all waves pooled.	24
4.14	Activity by self-rated health, long-term condition and mental-health condition.	24
4.15	Indoor vs outdoor activity over time, observed 2016–2022 and projected to 2025.	25
4.16	Indoor/outdoor preference by activity level, and where each activity level is active.	25
4.17	Inner vs outer London $\times$ socio-economic group, indoor and outdoor activity.	26
4.18	Volunteering in sport by background, pooled across waves. Described, not forecast.	26
4.19	Volunteer roles by demographic group, waves 2020–22.	27

List of Tables

3.1	Dataset summary	10
3.2	Distribution of the derived Activity_Level target variable.	11
3.3	Construction of the four broad age groups from the original nine-band age variable.	11
4.1	Comparison of predictive performance across the modelling approaches.	15
4.2	Decomposition of the near-term London-wide forecast into composition and behaviour effects.	19
4.3	Ten-year scenario range by activity/club type (% of London adults).	20

List of Algorithms

List of Listings

Abstract

Physical inactivity remains an important public health challenge across London, where participation in sport, physical activity and volunteering varies considerably between boroughs and demographic groups, contributing to existing health inequalities. Although the Active Lives Survey provides extensive information on participation and its demographic correlates, existing research has largely focused on describing historical patterns rather than forecasting how participation may change in the future. This project develops a decision-support framework for producing borough-level forecasts of participation across London. The project uses eight harmonised waves of the Active Lives Adult Survey from 2015–16 to 2022–23, covering respondents across all 32 London boroughs. The main challenge was the limited temporal information available for conventional forecasting. The dataset provides only 256 borough-year observations, with eight annual observations per borough and additional disruption associated with the COVID-19 pandemic. The project therefore separates the relationship between individual demographic characteristics and activity level from projected changes in population composition. An individual-level propensity model is combined with projected population characteristics to produce borough-level forecasts and distinguish changes associated with demographic composition from behavioural change.

Our research hypothesis is that separating a time-free demographic propensity model from a separately projected population will produce borough-level forecasts that are more statistically appropriate and interpretable for policy stakeholders than directly applying conventional forecasting models, without a substantial loss in predictive performance.

Main achievements:

- Harmonised eight Active Lives Survey waves, resolving changes in variable definitions, category coding and borough identifiers.
- • Developed an ordinal logistic regression propensity model and benchmarked it against LightGBM, achieving similar performance (AUC 0.704 and 0.707 respectively).
- Developed a forecasting framework combining the propensity model with projected population characteristics to distinguish demographic composition effects from behavioural change.
- Explored K-Modes with Random Forest, Bayesian hierarchical modelling and CatBoost with SHAP to inform and support the final methodology.
- Produced borough-level outputs covering overall participation, demographic differences, the indoor–outdoor balance and volunteering as a decision-support resource.

Overall, the findings support the proposed framework as a practical and interpretable alternative to directly applying conventional forecasting models to the limited borough-level time series available in the Active Lives data.

Summary of Changes

A conditional section, of at most 1 page

If (and only if) the dissertation represents a resubmission (e.g., as the result of a resit), then this section is compulsory: the content should summarise all non-trivial changes made to the initial submission. Otherwise you can omit it, since **a summary of this type is only needed for resubmissions**.

When included, the section will ideally be used to highlight additional work completed, and address criticism raised in any associated feedback. Clearly it is difficult to give generic advice about how to do so, but an example might be as follows:

- Feedback from the initial submission criticised the design and implementation of my genetic algorithm, stating “there seems to have been no attention to computational complexity during the design, and obvious methods of optimisation are missing within the resulting implementation”. Chapter 3 now includes a comprehensive analysis of the algorithm, in terms of both time and space. While I have not altered the algorithm itself, I have included a cache mechanism (also detailed in Chapter 3) that provides a significant improvement in average run-time.
- I added a feature in my implementation to allow automatic rather than manual selection of various parameters; the experimental results in Chapter 4 have been updated to reflect this.
- Questions after the presentation highlighted a range of related work that I had not considered: I have made a number of updates to Chapter 2, resolving this issue.

Supporting Technologies

A compulsory section, of at most 1 page

This section should present a detailed summary, in bullet point form, of any third-party resources (e.g., hardware and software components) used during the project. Use of such resources is always perfectly acceptable: the goal of this section is simply to be clear about how and where they are used, so that a clear assessment of your work can result. The content can focus on the project topic itself (rather, for example, than including “I used L^AT_EX to prepare my dissertation”); an example is as follows:

- I used the *Pandas* and *Seaborn* public-domain Python Libraries.
- I used a parts of the OpenCV computer vision library to capture images from a camera, and for various standard operations (e.g., threshold, edge detection).
- I used Amazon Web Services for remote storage and processing of data. Specifically, I used:
 - Simple Storage Service (S3) for data storage
 - Elastic Compute Cloud (EC2) for provision of virtual machines
 - Elastic Beanstalk for scaling and load management
 - Sagemaker for all the machine learning components of my project.
- I used L^AT_EX to format my thesis, via the online service *Overleaf*.

Notation and Acronyms

An optional section, of roughly 1 or 2 pages

Any well written document will introduce notation and acronyms before their use, *even if* they are standard in some way: this ensures any reader can understand the resulting self-contained content.

Said introduction can exist within the dissertation itself, wherever that is appropriate. For an acronym, this is typically achieved at the first point of use via “Advanced Encryption Standard (AES)” or similar, noting the capitalisation of relevant letters. However, it can be useful to include an additional, dedicated list at the start of the dissertation; the advantage of doing so is that you cannot mistakenly use an acronym before defining it. A limited example is as follows:

AES	:	Advanced Encryption Standard
DES	:	Data Encryption Standard
	:	
$\mathcal{H}(x)$	:	the Hamming weight of x
$\mathbb{F}_q$	:	a finite field with q elements
x_i	:	the i -th bit of some binary sequence x , st. $x_i \in \{0, 1\}$

Acknowledgements

An optional section, of at most 1 page

It is common practice (although totally optional) to acknowledge any third-party advice, contribution or influence you have found useful during your work. Examples include support from friends or family, the input of your Supervisor and/or Advisor, external organisations or persons who have supplied resources of some kind (e.g., funding, advice or time), and so on.

Dave Cliff writes here to say huge thanks to his colleague Dr Dan Page for sharing this L^AT_EX thesis template, which was originally written by Dan, for Computer Science dissertations. Dave edited Dan's original to better suit the needs of the Data Science MSc: please don't hassle Dan about any of this, but do feel free to contact Dave if you have any questions or comments on it.

Chapter 1

Introduction

1.1 Context and Problem Statement

Physical activity is one of the important factors that influences physical health, mental wellbeing and quality of life, though the level of participation varies significantly across various demographic groups and geographical areas, which contributes to chronic health inequalities and increased pressure on healthcare services.

London is an interesting city for researching participation due to the diverse population, socio-economic inequalities, and differences in access to sports facilities. Participation patterns vary across boroughs and different population segments, highlighting significant challenges in public health. These unequal levels of participation can contribute to wider health inequalities and also show that the benefits of being physically active are not distributed equally.

In order to effectively prepare for future changes, organisations and policymakers are eager to understand how this information can inform decisions regarding the allocation of resources and the implementation of health policies and drive impactful outcomes. General surveys of the country provide a large amount of information about historical participation and are primarily designed to describe what has happened rather than to predict what is likely to happen in the future.

This creates a practical problem for future participation: how can the available survey data be used to predict future participation when the data consist of repeated cross-sectional surveys rather than a panel of the same individuals over time? This predictive research problem is addressed by the current study, using data from 8 consecutive waves of the Active Lives Adult Survey (2015–16 to 2022–23), which includes data collected from participants in London that are spread across 32 London boroughs.

1.2 Motivation and Contribution

The significance of this project can be considered from both a practical and a methodological perspective. From a practical perspective, borough-level forecasts of participation broken down by demographic group and activity type can provide the council or other organisations with evidence to support target-setting, resource allocation and the planning of interventions, as these forecasts will help in identifying those demographic groups and activities for which the forecasted change in participation is most important.

The project also addresses a gap in the limitations of the existing literature. The majority of studies based on the Active Lives Survey have mainly used regression or multilevel model analysis to explore the relationship between demographic factors and physical activity. However, not enough work has been done so far to predict future participation based on such a dataset. This is a complicated task because of the nature of this dataset, as it is repeated cross-sectional and its questionnaire structure and variable definitions have also changed across all the annual waves. Therefore, the project explores a forecasting approach which is suited to these characteristics of the data.

Conventional time-series or machine-learning forecasting methods cannot be applied directly to this problem because the amount of temporal data is limited. There are only eight annual survey waves, and the data is also affected by changes in the survey structure and the disruption caused by the COVID-19 pandemic. This project separates the relationship between individual characteristics and activity level from changes in the future population structure, instead of relying on a model to learn a complex time trend from these limited observations. The relationship between demographic characteristics and activity level is estimated using the full survey sample, which provides a large number of individual observations.

Future population composition is projected separately and is used to estimate future participation. This approach also makes it possible to distinguish between changes caused by differences in demographic composition and changes in behaviour within demographic groups. This distinction is essential because an increase or decrease in overall participation may not necessarily mean that people's behaviour has changed, and it may partly result from changes in the characteristics of the population.

1.3 Central Challenges

The design of the project is influenced by three main challenges, and these were the deciding factors for how the data should be analysed and how the forecasting approach should be developed.

Harmonising a survey that changes over time. The Active Lives Survey is a repeated cross-sectional survey rather than a panel survey, which means that a new sample of respondents is collected in each annual wave. The structure of the survey also changed considerably across the eight waves used in this project. For example, the number of recorded variables increased from 292 to 601, while some variable definitions and local-authority identifiers changed between waves. The local-authority codes were affected by authority changes elsewhere in England, though the London boroughs themselves remained unchanged geographically. These differences needed to be resolved before the survey waves could be combined. Before the analytical stage, different members of the project team independently identified borough codes that had been incorrectly matched between waves. This highlighted the importance of carefully checking the geographic information before carrying out the later analysis, as such inconsistencies would silently corrupt every subsequent analytical stage.

Limited temporal information for conventional forecasting. The dataset used contains 32 boroughs across eight annual waves, which gives 256 borough-year observations. Three of these waves were also affected by the disruption associated with the COVID-19 pandemic. This information is relatively limited for a time-series or machine-learning model to learn reliable temporal patterns. Instead of trying to learn a complex time trend from these few observations, the project therefore separates the relationship between demographic characteristics and activity from the projection of future population composition. This allows the larger individual-level survey sample to be used when estimating the demographic relationship, while population changes are considered separately when producing the forecasts.

Maintaining interpretability for a non-technical audience. The forecasts are intended to support decision-making by local authorities and organisations around resource allocation and possible interventions. Therefore, interpretability was considered alongside predictive performance and computational efficiency. A more complex model is not necessarily more useful if its results are difficult for the intended audience to understand or justify. The final modelling approach was therefore selected based on a balance between predictive performance, interpretability and practical suitability. This consideration is further discussed in Chapter 2 and informed the comparison and selection of the final forecasting model.

1.4 Summary of Aims, Objectives and Achievements

Based on the challenges discussed above, the overall aim and objectives of this project are as follows:

Aim: To develop an interpretable borough-level forecasting framework for physical activity, sport and volunteering participation across London, using harmonised Active Lives Survey data from 2015–16 to 2022–23.

- **Objective 1:** To harmonise the eight Active Lives Survey waves into a consistent dataset by addressing inconsistencies in variable definitions, category coding and geographic coding.
- **Objective 2:** To develop a propensity model that estimates the relationship between individual demographic characteristics and activity level, and to compare its performance with a more flexible gradient-boosting benchmark.
- **Objective 3:** To investigate the extent to which the differences in participation between London boroughs can be explained by demographic composition and by borough-level differences.

- **Objective 4:** To develop a forecasting approach that combines the propensity model with projected future population characteristics while distinguishing between changes that are associated with population composition and changes in behaviour.
- **Objective 5:** To apply the framework to each of four areas: overall participation, demographic differences, the indoor–outdoor balance and volunteering. The results will then be presented in a form that can support decision-making by a non-technical stakeholder.

These objectives provide a structured approach for the development and evaluation of the borough-level forecasting framework. The analysis also aims to provide a clearer understanding of the factors associated with differences in participation between London boroughs and how these differences may affect future forecasts.

Chapter 2

Background

2.1 Problem Definition

Physical inactivity is one of the important and persistent public health challenges in London. Participation rates in sport and physical activity differ greatly between boroughs and different socio-economic groups, which could add to current health inequalities (Choe, Lee and Kang, 2025). The Active Lives Survey provides a comprehensive source of information for examining these differences. However, there are several characteristics of the survey that should be considered when it is used for forecasting. The survey is repeated cross-sectional, meaning that each annual wave contains a new sample of respondents rather than following the same individuals over time. In addition, the structure of the survey, variable definitions and category coding changed across the period covered by this project. Several of these changes were identified during the harmonisation of the 2015–16 to 2022–23 waves. For example, one socio-economic classification category was incorporated into an existing category from 2019–20 onwards. Similarly, the mid-level activity category was renamed from ‘Insufficiently Active’ to ‘Fairly Active’, although the underlying activity thresholds remained unchanged. These changes mean that these variables could not simply be combined across all waves without first evaluating whether their meanings were consistent.

Geographical identifiers provided another similar yet important issue. Local-authority codes are assigned across all English local authorities; if there is any change, such as mergers elsewhere in England, this can affect the numerical codes used for London boroughs in different survey waves. The London boroughs themselves have remained geographically fixed since 1965, but their numerical identifiers were not consistent across the survey files. Two members of the project team independently identified borough codes that had been incorrectly matched between waves using different verification procedures. This highlighted the potential for geographic coding errors to affect the analysis if they were not identified during the harmonisation stage. To address this issue, the local-authority codes from each wave were mapped using the relevant wave metadata to permanent Office for National Statistics Government Statistical Service identifiers. This provided a consistent geographic reference across the survey waves. These issues demonstrate that the forecasting problem cannot be considered solely in terms of selecting an appropriate modelling technique: the quality of the forecasting results also depends on whether the underlying survey data have been made sufficiently consistent across time. Careful harmonisation and validation were therefore treated as an important part of the modelling process rather than as a separate data-cleaning task.

2.2 Domain Background: Determinants of Participation

Various studies have investigated the number of factors that might affect engagement in physical activities. The studies of Brainard et al. (2019), Brinkley et al. (2020) and Davis et al. (2011) have suggested that factors such as age, gender, socio-economic position, geography and neighbourhood are important determinants of participation. Similar trends have been identified in the current project, including age, socio-economic classification, disability and ethnicity, which were proved to be significant predictors of activity band. Research using the Active Lives Survey has mainly focused on describing and explaining existing trends of participation. Regression analyses and multi-level methods have been used to analyse the relationships between demographic factors and physical activity. They help to understand what factors are related to participation but do not suggest a way to estimate changes in the future. This difference between explaining existing patterns and forecasting future participation is important for the

current project. Identification of the relationship between one demographic feature and physical activity does not necessarily mean that there is a way to make predictions about borough-level participation. The goal of the present research is to proceed from the existing knowledge of the factors affecting participation and use these relationships as part of a framework for estimating future participation at borough level.

2.3 Forecasting Approaches Considered and Limitations of Conventional Models

Several modern machine-learning forecasting methods have been developed for structured temporal data. These include Gradient Boosting Machines (Chen and Guestrin, 2016; Ke et al., 2017), the Temporal Fusion Transformer (Lim et al., 2021) and N-BEATS (Oreshkin et al., 2020). These approaches have been successful in areas such as finance, energy, and health care. One possible approach in this case would be to estimate the activity rate per year for each of the boroughs, whether through data of borough per year or fitting a time series for each borough. However, the available data provides limited information for this type of approach. The dataset consists of 32 boroughs in 8 waves annually, which means that there are 256 borough-year observations. In addition, 3 of the waves include the disruption of the COVID-19 pandemic. Although each borough-year estimate is based on approximately 530 survey respondents, there are still only eight observations over time for each borough. This is challenging for the conventional time-series model to detect meaningful changes over time from variation in the survey estimates. Classical forecasting methods such as Prophet and ARIMA were also considered and were not selected for the borough-forecasting analysis for similar reasons. A weighted linear trend was instead used as a more conservative approach in a supplementary analysis. These results supported the decision not to make the borough forecasting component dependent on a complex temporal model.

The borough-forecasting model focuses on the relationship between demographic characteristics and activity level rather than attempting to learn a detailed borough-level time trend. The model is estimated to use the full sample to capture the relationship between individual characteristics and activity band. Future population characteristics are then projected separately and combined with the estimated model to produce borough-level forecasts. This approach also makes it possible for forecasted participation changes to be examined from two different angles. Certain changes may be due to variations in the demographic makeup of the group, whereas other changes might occur as a consequence of behavioural changes on the part of people who have the same demographics. This separation offers greater insight than just looking at a single overall time-series forecast and is useful when considering how future participation could be influenced through policy or interventions.

2.4 The Borough-Forecasting Model

The outcome used in the borough-forecasting model is activity level, represented by three ordered categories: Inactive, Fairly Active and Active. As these activity levels can be categorised in an ordered manner, the ordered logit model was used which accounts for the ordering of the activities rather than assuming that they are independent of each other as in multinomial regression (McCullagh, 1980). The predictors included in the model were age band, gender, ethnicity, socio-economic classification, education, disability status, body-mass-index group and borough identity. Survey weights were also included during model fitting so that the estimates better reflect the population represented by the survey.

To finalise the main model, two different specifications were compared. Alongside the ordinal logistic regression, a gradient-boosting model using LightGBM (Ke et al., 2017) was also evaluated as a more flexible benchmark. The aim of this comparison was to check whether the introduction of a more complicated connection between predictors and activity level would result in some improvement in the prediction. The models were evaluated using a held-out survey wave. LightGBM achieved an AUC of 0.707, compared with 0.704 for the ordinal logistic regression. The difference in performance was small and was not practically meaningful. In addition to its similar predictive performance, it provides coefficients that can be interpreted directly and also makes the application of the model to future population characteristics easier. The comparison with LightGBM therefore provided the evidence that the additional complexity of a more flexible model did not provide a sufficient improvement in predictive performance to justify using it as the main model.

The model selection was also influenced by the limitations of the underlying data. Using a more complex algorithm does not remove the limitations created by having only eight annual survey waves and a repeated cross-sectional study design. The final model was therefore selected based on a combination of

predictive performance, interpretability and suitability for the forecasting problem rather than predictive accuracy alone. The model was then used to generate individual-level activity predictions, which were then combined with projected future population characteristics to produce the borough-level forecasts presented in later chapters.

2.5 Composition–Place Decomposition & Model Interpretability

An important question to consider for the purpose of this research is the extent to which differences in activity across boroughs are due to individual characteristics of the people or to characteristics of the borough itself. This issue is significant for the forecasting process since projections for the future will take into account the change in demographic structure whereas characteristics of the place cannot be projected in the same way. To investigate this, three model specifications were compared: a demographics-only model, a borough-only model and a model including both demographic characteristics and borough identity. These models were compared based on how well they reproduced the observed variation in activity between boroughs. The demographics-only model was also evaluated using a leave-one-borough-out procedure, where each borough was predicted using a model trained without observations from that borough.

The demographics-only model explained 48% of the variation between boroughs. When borough identity was included alongside the demographic variables, this increased to 88%. This suggests that demographic composition accounts for a substantial part of the differences in participation between boroughs, while borough identity provides additional information. The leave-one-borough-out result was 47%, which was very similar to the 48% obtained from the corresponding demographics-only model. This provided evidence that the relationship between demographic characteristics and activity level is reasonably consistent across boroughs, including those that were not used to train the model. This is relevant to the forecasting approach because the model will be applied to projected population characteristics in the future. The composition-place method is connected with the demographic decomposition approach developed by Kitagawa (1955), where group differences are explained through differences in composition and differences in rates among the groups. This helps us to determine whether participation differences are due to the demographic make-up of the borough population or due to the borough itself.

Interpretability also plays a major role in choosing the primary method Molnar (2022). Both the ordinal regression and the composition-place analysis methods have interpretable coefficients that can be interpreted directly. For that reason, post-hoc interpretation tools such as SHAP (Lundberg and Lee, 2017) and variable importance using permutation (Fisher et al., 2019) were not applied to the primary model. Such methods are helpful, especially if the underlying model is not straightforward to interpret. SHAP was also applied separately in the gradient boosting analysis that was a part of the wider project which is discussed in Section 2.6. The results of the analysis serve as additional support for the conclusions rather than the primary method of interpreting the model.

2.6 Alternative Approaches Explored

In addition to the borough-forecasting model described above, three alternative modelling approaches were undertaken as separate, independent workstreams within the wider project outlined in Chapter 3. Each approach addressed a distinct question in its own right and was developed and evaluated on its own terms.

Demographic clustering and activity-level classification. One approach combined unsupervised clustering of respondents based on demographic characteristics with supervised classification of overall activity level. K-Modes was used for the clustering because the main demographic variables, such as age band, gender and ethnicity, were categorical (Huang, 1997). Random Forest was then used to classify respondents according to their activity level (Breiman, 2001). This approach produced useful demographic segments and helped identify which characteristics were most informative for predicting activity level. However, since a CatBoost model was already being used as a separate approach for individual-level activity classification, this clustering-and-classification approach was found to be largely redundant. Its main contribution to the final methodology was the identification of demographic variables that were also considered in the propensity model.

Bayesian hierarchical modelling. A second approach which was considered is the hierarchical structure of the survey data, with individuals nested within boroughs and boroughs within London. A Bayesian hierarchical model was developed with demographic characteristics as fixed effects and boroughs were included as a random effect by following the general Bayesian workflow Gelman et al. (2020) and the approach to grouped data discussed by Lunn et al. (2013). The models were fitted using Markov Chain Monte Carlo (MCMC) through PyMC/Stan. Individual model runs took approximately five to six hours, which limited the amount of iterations that could be carried out during the development. Despite this limitation, the analysis provided evidence that borough identity contains information beyond demographic composition. This finding was also examined through the composition-place analysis.

Gradient-boosted modelling, SHAP and borough residual analysis. A third workstream used a CatBoost gradient-boosting classifier for the individual-level prediction task. The model was trained on waves one to seven and evaluated on the final wave. This temporal validation approach was preferred over a random train-test split because it reduces the risk of using information from a later wave when predicting an earlier stage of the analysis. The model achieved an AUC score of 0.708, which was very similar to the performance obtained from the models considered in the primary pipeline. SHAP analysis (Lundberg and Lee, 2017) was used to investigate the importance of the predictors. Age was the most influential variable, which was followed by ethnicity, education and social grade. Using this pipeline, we also examined borough-level residuals using a leave-one-wave-out procedure. Nine boroughs showed a stable, sign-consistent effect across at least seven of the eight waves. This effect remained when controlling for area deprivation, with a correlation of 0.06. These results provided additional evidence that differences between boroughs may not be explained entirely by demographic composition.

An additional borough-level forecast was also developed using a weighted linear trend. This approach was chosen after some time-series methods (Prophet, ARIMA), which are more flexible but were considered unsuitable due to the short length of only eight annual years. Another separate activity-type analysis examined the contribution of individual activities to the 150-minute weekly activity threshold and showed that walking and active travel were responsible for a significant part of the activity rate of Londoners. However, numerous valuable insights were gained through this approach; it was not chosen as the primary forecasting method as it did not provide any distinguish between demographic composition and behavioural change and SHAP analysis gave data which could be easily obtained via interpretable coefficients of ordinal regression. Although, borough residual and activity-level analysis were helpful in corroborating insights gained from the main method of forecasting.

2.7 Related Work and Research Gap

Another study employed various machine learning techniques to predict adherence to physical activity guidelines based on data collected from cross-sectional surveys (Choe, Lee and Kang, 2025). Six algorithms were tested, and the results revealed that features such as sedentary behaviour, age, gender and education were most useful for prediction. This shows that machine learning techniques could be applied to predict physical activity outcomes at an individual level. However, that study focuses on current adherence rather than changes in the future and does not provide geographically disaggregated forecasts. The literature and methodological considerations reviewed in this chapter reveal four gaps that the present project addresses.

1. While repeated cross-sectional surveys can provide a large number of individual observations, they may contain changes in questionnaire structure, variable definitions and coding between waves. Additional challenges can arise when the data is used for forecasting rather than simply describing historical patterns.
2. An application gap exists in existing research using the Active Lives Survey. Most studies have mainly focused on describing levels of participation or investigating relationships between demographic characteristics and activity, and fewer attempts have been made to use the survey to produce future participation forecasts.
3. Methods such as Gradient Boosting Machines, the Temporal Fusion Transformer and N-BEATS can be effective when sufficient temporal data is available. However, the Active Lives data contain only eight annual waves at borough level, which means that a complicated forecasting model cannot learn reliable temporal patterns specific to each borough. Instead, the method proposed in this project

leverages the existing individual-level data to estimate a time-free relationship between demographic characteristics and activity, and then combines this with future population projections.

4. There is limited literature that combines harmonised multi-year Active Lives data with borough-level forecasting while considering the possibility of future changes due to demographic differences or behavioural change. This is an important gap to address because a change in participation does not necessarily imply a change in the behavioural pattern of the borough, as it may result from changes in the population structure of a borough.

These gaps motivate the methodology described in subsequent chapters. The project begins with the harmonisation of the survey data, followed by the development and validation of the propensity model, examines the association between demographics and borough, and uses these components within the forecasting framework.

Chapter 3

Methodology

3.1 Overall Project Workflow

This project followed a structured five-stage machine learning pipeline, transforming raw Active Lives Survey data into borough-level forecasts across four analytical pillars: *P1* overall physical activity participation, *P2* demographic-specific participation, *P3* indoor versus outdoor activity, and *P4* sport volunteering. The workflow was designed to ensure that data quality issues were addressed before modelling, enabling the development of reliable, interpretable and generalisable predictive models.

Stage 1 – Data Preparation and Harmonisation: eight structurally inconsistent annual survey files were merged into a single individual-level dataset. Preprocessing was required to standardise variable names, harmonise borough identifiers using permanent ONS GSS codes, resolve inconsistencies in Local Authority coding, and handle sentinel/missing values.

Stage 2 – Exploratory Data Analysis: a quality-assurance EDA targeting six specific risk areas associated with a multi-wave government survey: merge integrity, sentinel contamination, missingness, borough coverage, cross-year consistency and distributional plausibility.

Stage 3 – Feature Engineering: the cleaned master dataset, containing 610 variables, was reduced to a compact model-ready dataset, with the ordinal `Activity_Level` target derived and demographic predictors consolidated.

Stage 4 – Machine Learning Modelling: our separate modelling approaches were developed and evaluated: CatBoost with SHAP for interpreting individual-level predictors, LightGBM for borough-level forecasting, K-Modes clustering with supervised classifiers for demographic segmentation, and a Bayesian Hierarchical Ordinal model for assessing borough-level variation and uncertainty.

Stage 5 – Forecasting and Scenario Analysis: combines model outputs with projected demographic composition and behavioural trends to produce borough- and London-level forecasts. Changes are decomposed into demographic and behavioural effects, while activity-type analysis identifies walking as the main contributor to the 150-minute threshold and highlights differences in activity patterns across boroughs.

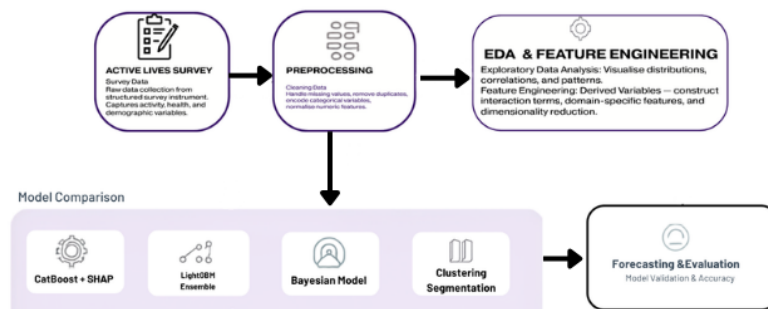


Figure 3.1: Overall project workflow diagram.

A consistent validation principle was applied across all four modelling strands: rather than using random train–test splits, a time-based validation framework was used. Models were trained using the first seven years of data and tested on the final 2022–23 survey wave. This approach closely reflects the real-world forecasting problem by ensuring that predictions are generated using only historical information, thereby preventing information leakage and providing a more realistic assessment of model generalisation.

3.2 Dataset Description

The analysis uses data from Sport England’s Active Lives Adult Survey, a nationally representative survey that measures physical activity participation among adults aged 16 years and above in England. The survey collects information on physical activity behaviour, demographic characteristics, health, education, volunteering and other lifestyle factors using a repeated cross-sectional design. Unlike a longitudinal study, respondents are not followed over multiple years; instead, a new sample is collected annually.

Eight consecutive survey waves, spanning 2015–16 to 2022–23, were used. After filtering the national dataset to Greater London and excluding the City of London, the final dataset comprised 135,497 respondents across 32 London boroughs. Although each raw survey contained approximately 10,700 variables, only variables relevant to the four analytical pillars were retained. As additional question blocks were introduced in later survey waves, the number of columns increased from 292 in the first wave to 601 in the final wave. Table ?? summarises the key properties of the dataset.

Property	Value
Data source	Sport England Active Lives Adult Survey (UKDS Ref. 9288)
Survey period	2015–16 to 2022–23
Survey waves	8
Study area	32 London boroughs
Total respondents	135,497
Raw variables (per wave)	~10,700
Merged dataset variables	610

Table 3.1: Dataset summary

3.3 Data Preprocessing

Before model development, a single consistent analytical dataset was required. The Active Lives Survey evolved over time, with different geographical coding, variable availability, naming conventions and survey structure. During preprocessing the focus was on harmonising inconsistencies while preserving the integrity of the original survey data. The overall preprocessing workflow is illustrated in Figure 3.2.

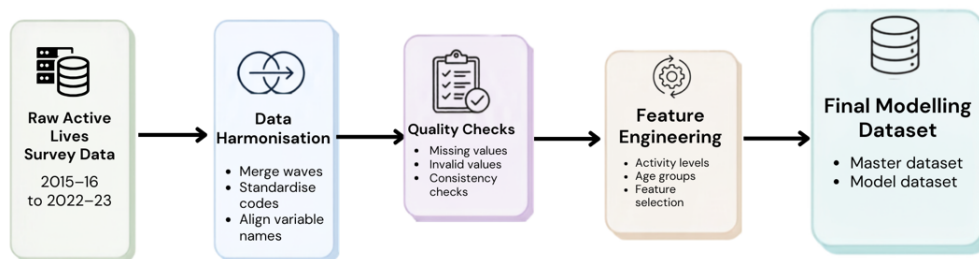


Figure 3.2: Data preprocessing workflow showing dataset harmonisation, data quality checks, feature engineering, and creation of the final modelling datasets.

3.3.1 Data Harmonisation

The first stage was harmonising the eight survey waves into a unified dataset. Borough identities were standardised using the permanent **ONS Government Statistical Service (GSS)** codes to ensure

that each London borough remained consistently identified across all survey years. Variable names were also standardised where naming conventions had changed between survey waves, to merge equivalent variables into a common schema. These steps ensured that the final merged dataset contained consistent geographical and variable identifiers across the entire study period.

3.3.2 Missing Data and Data Quality

Data quality checks were performed to identify missing values, invalid entries and inconsistencies across survey waves. During import it was essential to convert the SPSS sentinel values representing non-response into missing values, to prevent invalid numeric codes from influencing the prediction. The remaining missing values arose primarily from survey routing or from questions introduced in later survey waves, rather than from data corruption.

Rather than imputing missing values, the original survey structure was preserved wherever possible. Some variables, such as health, were unavailable in earlier survey years and remained missing for those years. Optional demographic non-response categories were retained as explicit categories to avoid unnecessary information loss. Minor inconsistencies in categorical labels across survey years were standardised to ensure a consistent coding framework throughout the merged dataset.

3.3.3 Feature Engineering

Feature engineering transformed the harmonised dataset into a model-ready format. The primary target variable, **Activity_Level**, was derived from the variable **MEMS7_ALL** (number of minutes of moderate-intensity activity per week): Inactive (0–29 minutes/week), Fairly Active (30–149 minutes/week), and Active (150 or more minutes/week). The resulting target variable was stored as an ordered categorical variable to preserve its ordinal structure for modelling. The resulting distribution of the target variable is shown in Table ??.

Activity Level	MEMS7_ALL Threshold	Count	% of Dataset
Active	150+ minutes/week	90,124	66.5%
Fairly Active	30–149 minutes/week	15,033	11.1%
Inactive	0–29 minutes/week	30,340	22.4%

Table 3.2: Distribution of the derived **Activity_Level** target variable.

To improve interpretability and maintain adequate sample sizes, the age variable, originally recorded using nine age categories, was redefined into four broad age groups (16–24, 25–44, 45–64 and 65+), as detailed in Table ?. All categorical predictors were converted to categorical data types to improve computational efficiency and ensure compatibility with the machine learning algorithms.

Age_Group	Age9 Bands Included	Count
16–24	16–24	11,110
25–44	25–34, 35–44	53,216
45–64	45–54, 55–64	43,555
65+	65–74, 75–84, 85+	26,289

Table 3.3: Construction of the four broad age groups from the original nine-band age variable.

3.3.4 Final Modelling Dataset

The final master dataset contained 135,497 respondents and 610 variables. To improve modelling efficiency, two purpose-specific datasets were created. For the CatBoost, clustering and Bayesian models, a compact 20-variable dataset (containing demographic characteristics, activity measures and the derived target variable) was used. For activity-type information, a broader dataset with approximately 75 variables was used in the LightGBM model. Additionally, borough-level aggregated datasets were created to support trend analysis and borough-level forecasting, although these were not used for model training.

3.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted after preprocessing to understand the characteristics of the harmonised dataset and identify patterns that could influence model development. The EDA was carried out in two stages. First, each survey wave was analysed individually to examine borough coverage, demographic distributions and data completeness before merging. The second stage was a weighted EDA conducted on the harmonised single dataset, to explore London-wide demographic, geographical and temporal patterns. This approach ensured that any wave-specific inconsistencies were identified before they could influence the merged dataset.

The weighted exploratory analysis had four key objectives: respondents' demographic characteristics, trends in physical activity by year, geographical variation across boroughs, and volunteering behaviour. These analyses provided a better understanding of the study population, helped identify important patterns in the data, and informed the feature selection and modelling approaches used in the later stages of the project.

3.4.1 Population Characteristics

The demographic characteristics of the study population were explored using population-weighted summaries of age, gender, ethnicity, education and disability status, as shown in Figure 3.3. Examining these distributions helped establish an overall profile of the London population represented in the survey, and showed clear differences across demographic groups, supporting their use as important predictor variables in the subsequent modelling.

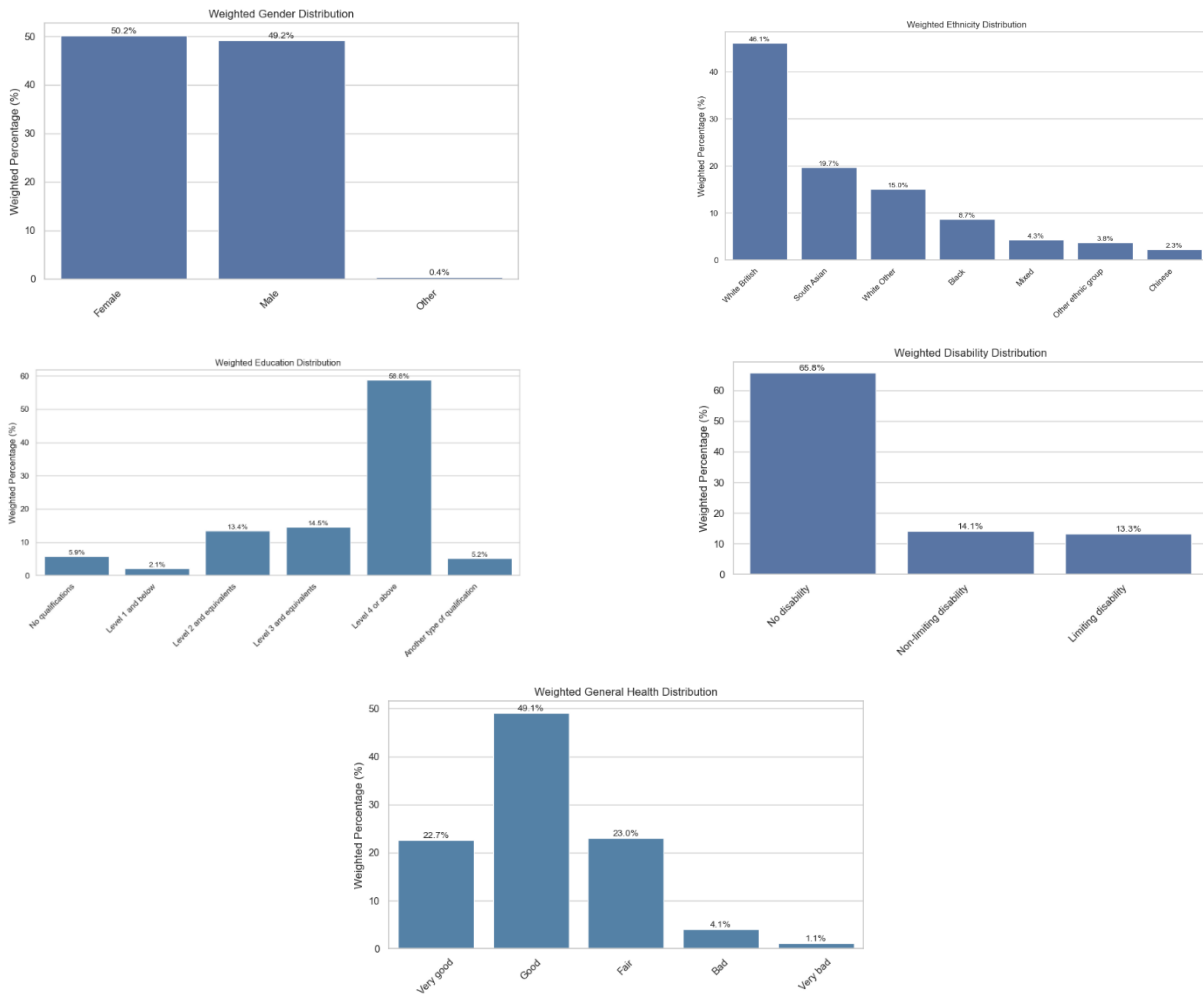


Figure 3.3: Population-weighted demographic characteristics of respondents.

3.4.2 Physical Activity Trends

Physical activity levels were analysed across all eight survey waves using the derived `Activity_Level` variable, as shown in Figure 3.4. Throughout the study period overall participation was stable; however, there was a temporary spike in inactivity during the COVID-19 pandemic, followed by a gradual recovery in later years. The proportion of respondents classified as Fairly Active also showed a gradual decline in later years, suggesting changes in participation behaviour over time.

Spatial analysis, shown in Figure 3.5, further demonstrated that activity levels varied across London's boroughs, indicating that geographical location may influence physical activity participation. These observations supported the inclusion of borough-level effects within the forecasting and Bayesian modelling frameworks.

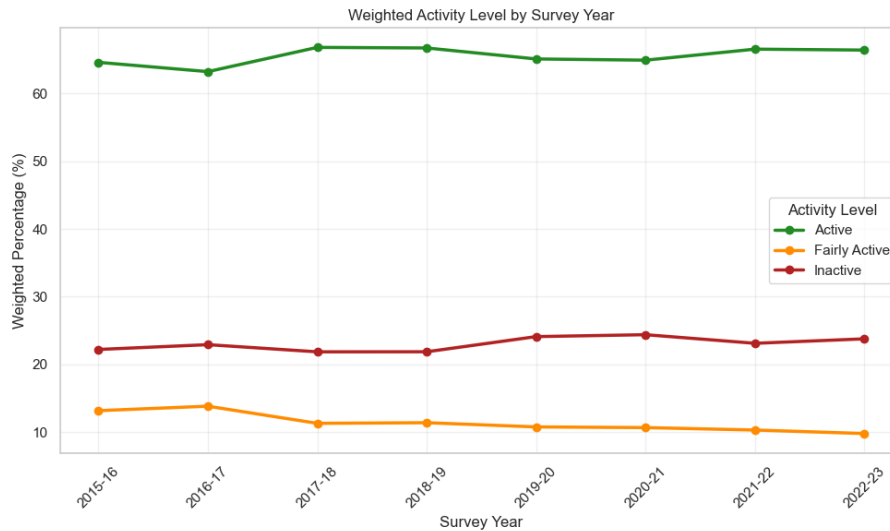


Figure 3.4: Population-weighted distribution of activity levels across survey years.

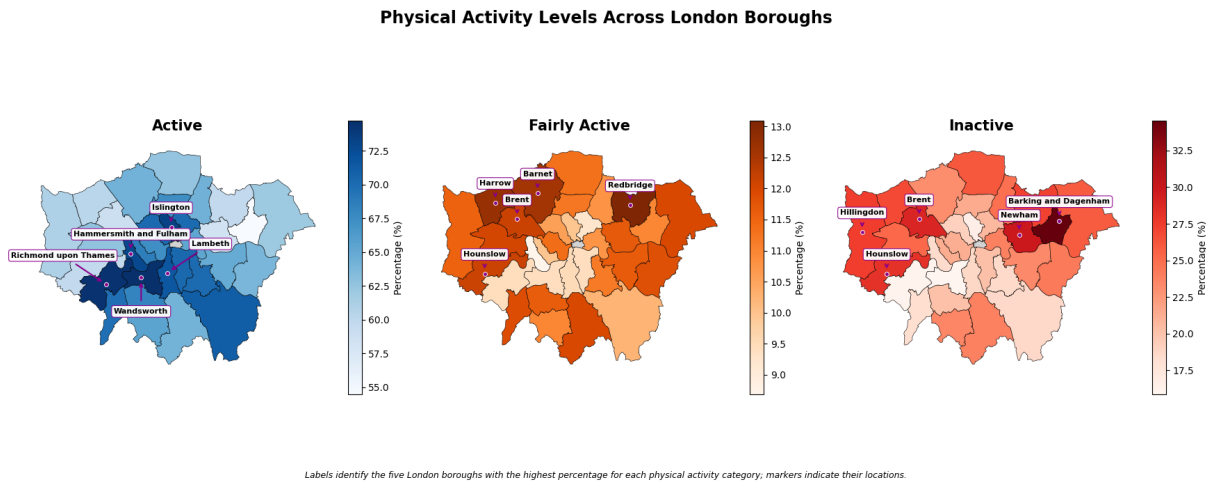


Figure 3.5: Geographical variation in population-weighted activity levels across London's boroughs.

3.4.3 Volunteering Trends

Volunteering was analysed to understand how participation changed over time, as shown in Figure 3.6. The results showed that volunteering levels fell during the COVID-19 period before gradually recovering in the following survey years. A noticeable change between the 2017–18 and 2018–19 survey waves was also observed, which aligns with revisions made to the Active Lives Survey rather than reflecting a genuine change in volunteering behaviour. Identifying these patterns helped provide context for the forecasting analysis presented in later chapters.

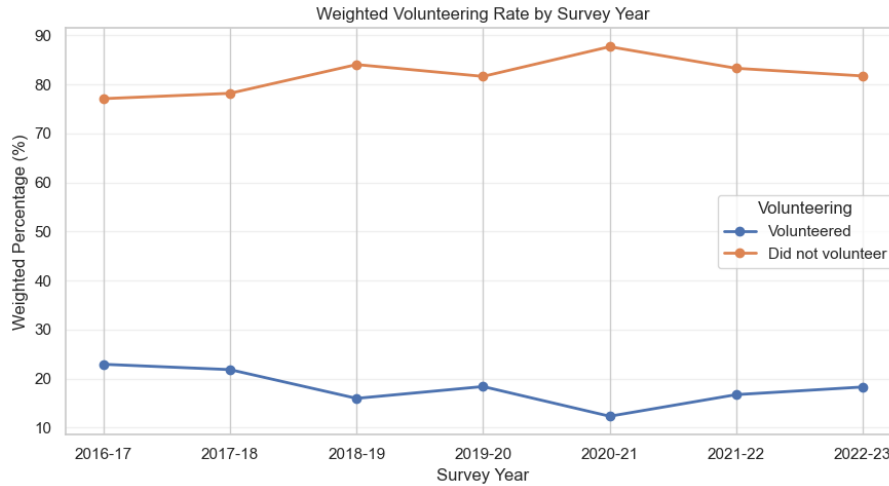


Figure 3.6: Population-weighted volunteering participation across survey years.

3.5 Machine Learning Framework

To address the objectives of the London Sport project, four different machine learning approaches were developed and evaluated independently. Rather than relying on a single model, multiple techniques were explored because each model offers different strengths in analysing physical activity data. Some models are better suited to prediction, others provide greater interpretability or uncertainty estimation, while some are useful for identifying hidden patterns within the population. Using a range of approaches also allowed their performance to be compared under the same data and validation framework before selecting the most suitable models for forecasting and interpretation.

The four modelling approaches included CatBoost with SHAP, LightGBM, K-Modes clustering, and a Bayesian Hierarchical Ordinal Logistic Regression model. CatBoost was selected for its strong performance on structured datasets containing many categorical variables, while SHAP was used to explain the contribution of each predictor to the model’s decisions. LightGBM was chosen as an alternative gradient-boosting approach because of its computational efficiency and ability to handle high-dimensional tabular data. K-Modes clustering was applied to identify groups of respondents with similar demographic and participation characteristics, giving additional insight into population segmentation. Finally, the Bayesian hierarchical model was developed to account for the ordered nature of the activity outcome while capturing variation between London boroughs through random effects and providing probabilistic estimates of uncertainty.

To ensure a fair comparison, all four models were developed using the same modelling dataset and evaluated under the same validation strategy. Survey data from 2015–16 to 2021–22 were used for model development, while the final 2022–23 survey wave was used as an unseen test dataset.

Chapter 4

Results and Critical Evaluation

4.1 Experimental Setup

All the models were evaluated on the same pooled dataset (135,497 respondents, 8 waves, 32 boroughs), but under two slightly different temporal validation strategies designed to evaluate performance on an unseen future survey wave. CatBoost and LightGBM were trained on waves 1–6, tuned on wave 7, and tested on the untouched wave 8 (2022–23). The Bayesian model and the clustering-based classifiers were trained on waves 1–7 as a single block (no separate tuning wave) and tested on wave 8. All splits are temporal rather than random; this approach mirrors the forecasting setting and avoids information leakage.

Not every model was evaluated on identical metrics or an identical target. CatBoost and clustering predict a **binary** outcome (meets the 150-minute guideline or not); LightGBM and the Bayesian model predict the **3-band ordinal** `Activity_Level`. The evaluation for supervised predictive models was carried out using **AUC**, **Log Loss** and **Accuracy**, while **runtime** was used to assess computational efficiency. The clustering and Bayesian approaches were assessed using measures appropriate to their respective modelling objectives, together with runtime. The results were considered in terms of predictive performance, computational requirements, interpretability and suitability for the project’s forecasting objective.

4.2 Model Comparison – Performance and Critical Analysis

The four approaches produced different strengths and limitations, reflecting their different modelling objectives. Table 4.1 summarises the metrics available for each approach.

Model	Target	AUC	Log Loss	Accuracy	Macro F1	Runtime
LightGBM	3-band ordinal	0.707	–	67.4%*	–	15 min 50 sec
CatBoost	Binary	0.708	0.588	68.9%	–	2 min 30. sec
Random Forest (clustering)	Binary	–	–	54.0%	0.428	14 sec
Gradient Boosting (clustering)	Binary	–	–	67.9%	0.363	2 min 16 sec
Bayesian Hierarchical Ordinal [†]	3-band ordinal	0.680	0.820	65.0%	0.570	~96 min

Table 4.1: Comparison of predictive performance across the modelling approaches.

Notes: * Weighted accuracy. † Bayesian results are based on a preliminary run using 200 posterior draws, 200 tuning iterations, and 2 chains; therefore, its metrics and runtime are not directly comparable with the other models and are included for indicative comparison only.

4.2.1 LightGBM

LightGBM achieved an AUC of 0.707, weighted accuracy of 67.4% and borough correlation of 0.90. It was marginally ahead of a benchmarked ordinal logistic regression (AUC 0.704, accuracy 67.1%, correlation 0.90) on every metric tested, as shown in Figure 4.1. The model provided a useful balance between predictive performance and computational efficiency, completing the full modelling process in approximately **15 minutes and 50 seconds**. It was also able to model non-linear relationships within the structured survey data and could be incorporated directly into the borough-level forecasting framework. LightGBM was therefore well suited to the forecasting stage, where both predictive performance and the ability to incorporate a broader set of activity-related features were important.

Model comparison on the untouched holdout wave — grey = no-features reference

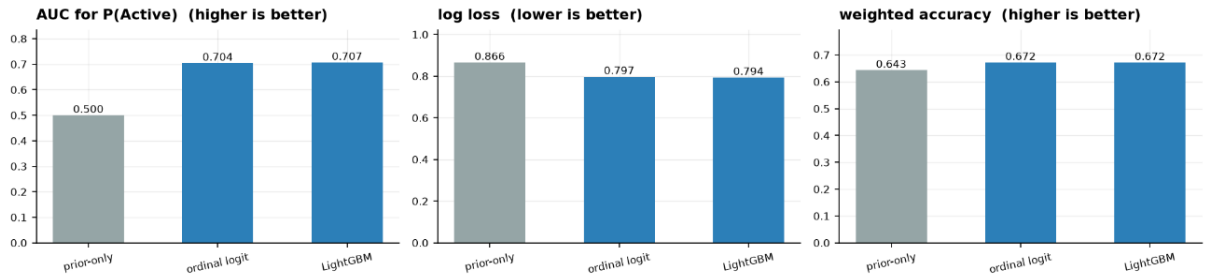


Figure 4.1: LightGBM predictive performance relative to the benchmarked ordinal logistic regression.

The model showed limited evidence of overfitting. Performance remained stable on unseen data, while the leave-one-borough-out check produced a borough-level correlation of 0.898, very close to 0.900 when boroughs were included. The borough-level correlation and calibration reliability plot, shown in Figure 4.2, also indicate that predicted probabilities broadly followed the observed probabilities, supporting the stability of the model's predictions.

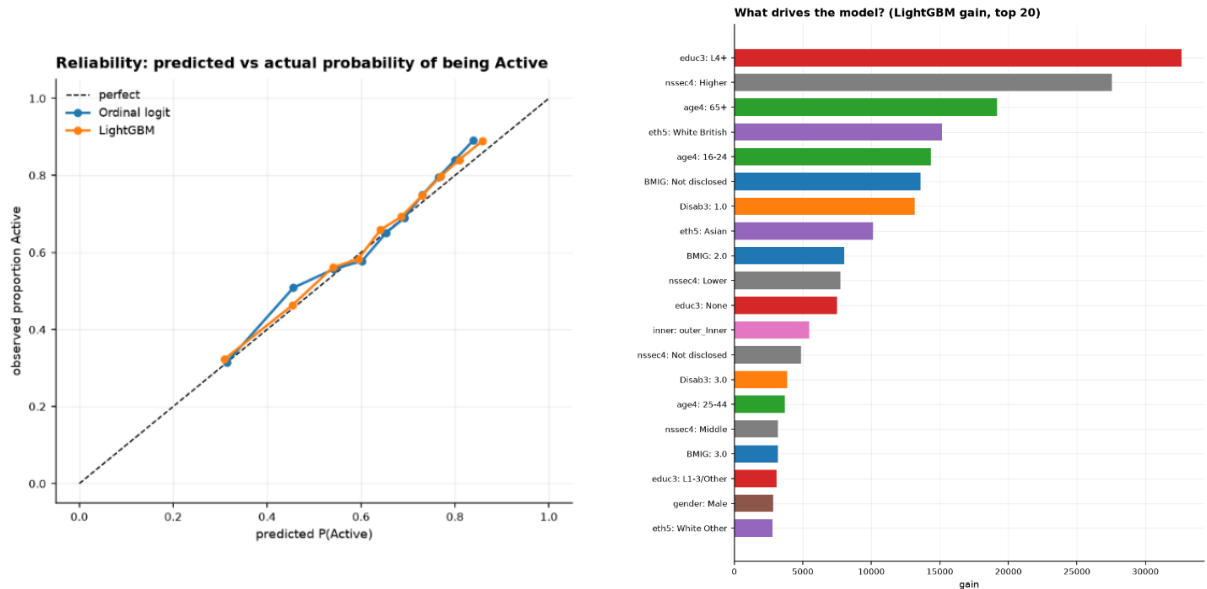


Figure 4.2: LightGBM borough-level correlation (left) and calibration reliability plot (right).

Feature importance indicates that education and socio-economic status were the strongest contributors to the model's predictions, followed by age. Ethnicity, disability and BMI also contributed meaningfully, while gender had a smaller influence. This suggests that socioeconomic and demographic characteristics played a greater role in predicting activity level than gender alone.

4.2.2 CatBoost + SHAP

CatBoost achieved an **AUC of 0.708**, **Log Loss of 0.588**, and **68.9% accuracy**. The main value of the model was its interpretability through SHAP. **Ethnicity, education and BMI** had the highest average influence, followed by age, social grade, disability and gender. Age showed the widest range of individual effects, while gender had the lowest overall influence. An interesting finding was that **missing BMI values were strongly associated with inactivity**, suggesting that the missingness itself contained useful information; this represents a model-based association rather than a causal relationship.

CatBoost provided strong individual-level prediction and useful insights into the factors associated with activity participation. However, its **AUC of 0.708** also indicates that demographic characteristics alone have limited ability to predict individual behaviour. The model also showed a small calibration shift on the held-out wave, with predicted probabilities approximately **1 percentage point below the observed values**, and tree-based models cannot naturally extrapolate to future periods. Therefore, CatBoost was retained primarily as an **interpretability and predictive benchmark**, with SHAP providing useful insights into the demographic drivers, rather than being used as the final forecasting model.

4.2.3 Clustering (K-Modes + Classifiers)

The K-Modes clustering approach was used to identify population segments based on categorical demographic and activity characteristics. The resulting clusters were subsequently evaluated using supervised classifiers. Gradient Boosting had the highest accuracy (67.9% test, 68.1% CV) but the weakest Macro F1 (0.363); Random Forest gave the best balance of accuracy and Macro F1 (54.0% / 0.428) and was selected within this strand. Its advantages were an appropriate categorical clustering method, interpretable population segments, and an empirical multi-model, multi-metric comparison. However, it uses a binary target and excludes survey year entirely, so it cannot represent temporal change or generate forecasts; this is why the whole strand was retained as exploratory rather than adopted as the final methodology.

4.2.4 Bayesian Hierarchical Ordinal Model

The Bayesian hierarchical ordinal model gave a useful alternative by treating Activity Level as an ordered outcome and allowing for differences between boroughs. It achieved an AUC of 0.68, accuracy of 65%, Macro F1 of 0.57 and Log Loss of 0.82. While these results were slightly below the supervised models, the Bayesian model provided useful information about borough-level variation and uncertainty. The borough random-effect standard deviation was approximately 0.123, suggesting modest differences between boroughs after accounting for the other predictors. However, the model took around 96 minutes to run, with some R-hat values around 1.03 and maximum tree-depth warnings. Overall, it was useful for interpretation, but less practical for repeated forecasting.

4.2.5 Final Model Selection: LightGBM

LightGBM was chosen as the main forecasting model because it gave the best balance between performance, speed and practical use. It achieved an AUC of 0.707 and accuracy of 67.4%, while taking only around 16 minutes to train. It can also capture non-linear relationships and interactions between factors such as demographics, year and borough without requiring them to be manually specified. This makes it well suited to the complex patterns in physical activity and to generating forecasts across multiple boroughs. CatBoost gave similar predictive results and was useful for SHAP-based interpretation, while the Bayesian model provided useful uncertainty and borough-level insights but was much slower. Overall, LightGBM was therefore selected for the final forecasting stage.

4.3 Physical Activity and Volunteering Trends Over the Next Ten Years

This section reports the forecast for all four analytical pillars: overall participation forecasts, demographic-specific insights, indoor versus outdoor activity, and volunteering in sport.

A note on forecasting versus description

Forecasting was applied only where the data provided sufficient evidence for reliable projection: the question had to be measured consistently across survey years, have adequate sample coverage, and show sufficient predictive structure from demographic variables. Overall activity, activity types, club membership and key demographic groups met all three criteria and were therefore forecast, with borough-level projections and quantified uncertainty.

Where these conditions were not met, results are presented descriptively rather than as forecasts, so that no figure implies more confidence than the data can support. Free time was asked only in the final wave, so there is no trend to project and it can only be described as a current snapshot. Volunteering was measured inconsistently, as its question base changed twice during the series, and demographics explain only around 15% of the variation between boroughs – too weak a signal to drive a projection, so it is reported at London and demographic level rather than forecast. Indoor and outdoor activity was asked of just ~30% of respondents, a rotating subsample that leaves the estimates with considerably greater uncertainty, so it is described with only one cautious London-level projection. Health was unavailable in the early survey years, leaving only five years of data, two of them affected by COVID and excluded as atypical; the remaining three clean years are not sufficient to establish a reliable trend, so despite being the strongest single correlate of activity, it too is reported descriptively.

This distinction ensures that forecasts are supported by the available evidence rather than presenting uncertain estimates as precise future outcomes.

4.3.1 P1 – Overall Participation Forecasts

Physical activity across London as a whole. The model projects broad stability in London’s overall activity levels in the near term. By 2025–26, around 64.9% of adults are projected to be Active, while 26.0% are Inactive and 9.1% are Fairly Active, as shown in Figure 4.3. Although the Active share changes little, the Fairly Active group has declined steadily over the observed period, suggesting a gradual shift away from the middle activity band.

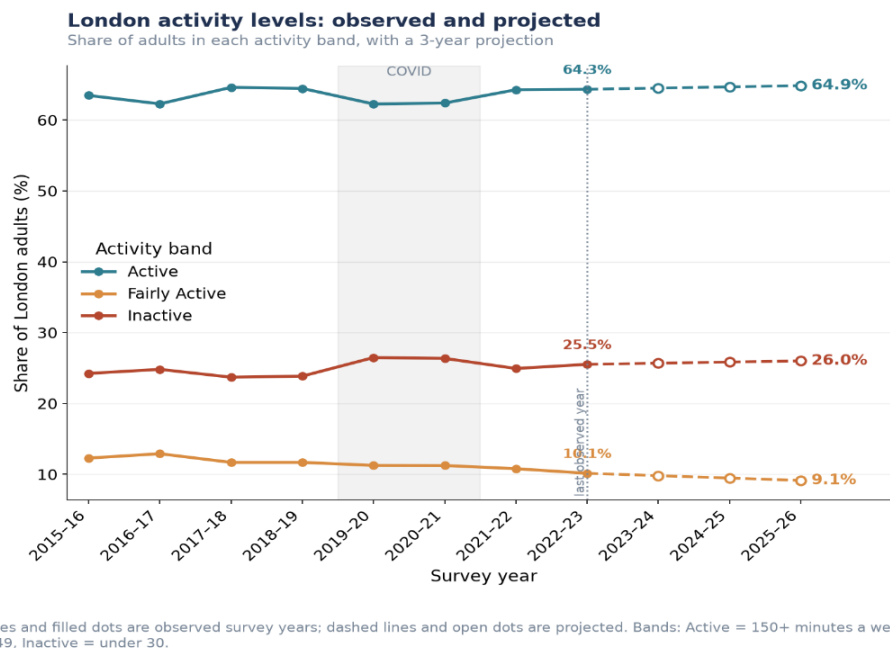


Figure 4.3: London-wide activity bands, observed 2015–16 to 2022–23 and projected to 2025–26.

Component	Effect by 2025–26	Interpretation
Composition	−0.37 points	London’s ageing population pulls activity down
Behaviour	+0.69 points	People of a given type are becoming more active
Net	+0.32 points	Apparent stability produced by two real trends cancelling

Table 4.2: Decomposition of the near-term London-wide forecast into composition and behaviour effects.

This decomposition, given in Table 4.2, is the central analytical result behind the ‘stable’ headline. A conventional forecast would report the number as roughly flat and stop there. Decomposing it shows that the stability is a coincidence of two genuine trends cancelling, not a settled equilibrium. If the behavioural improvement stalls, the demographic headwind takes over – the stability is more fragile than the flat line suggests.

For the longer-term outlook, three scenarios are presented rather than a single point forecast, as shown in Figure 4.4. By 2031–32, the Active share ranges from approximately 61.0% under a trend-reversal scenario to 65.4% if recent gains continue, with a population-change-only scenario at around 63.1%. This range highlights that the long-term direction of physical activity is sensitive to the assumptions made about future behaviour.

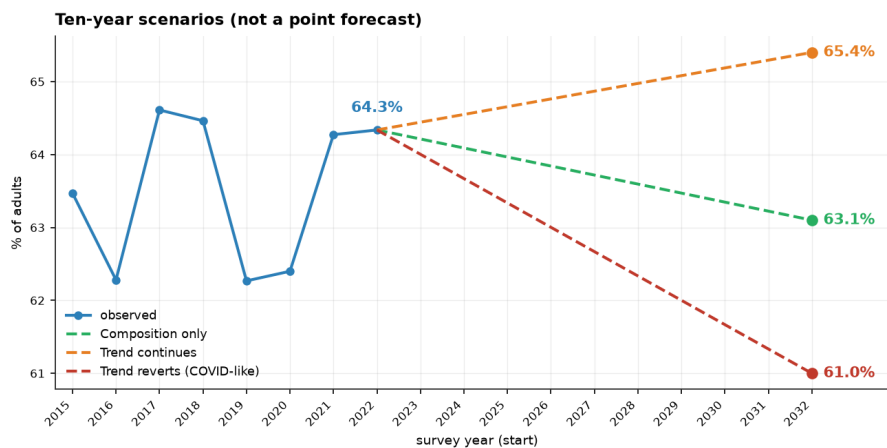


Figure 4.4: Ten-year scenario range for London-wide activity, 2022–2032.

Likelihood of participating in specific activities. Ten-year scenario and three-year scenario ranges were produced for seven individual activities and three club-membership measures. Among these, the top two activities are walking (Figure 4.6) and active travel (Figure 4.5); club membership trends are shown in Figure 4.7.



Figure 4.5: Active travel: ten-year scenarios (widest scenario spread of any activity measured).

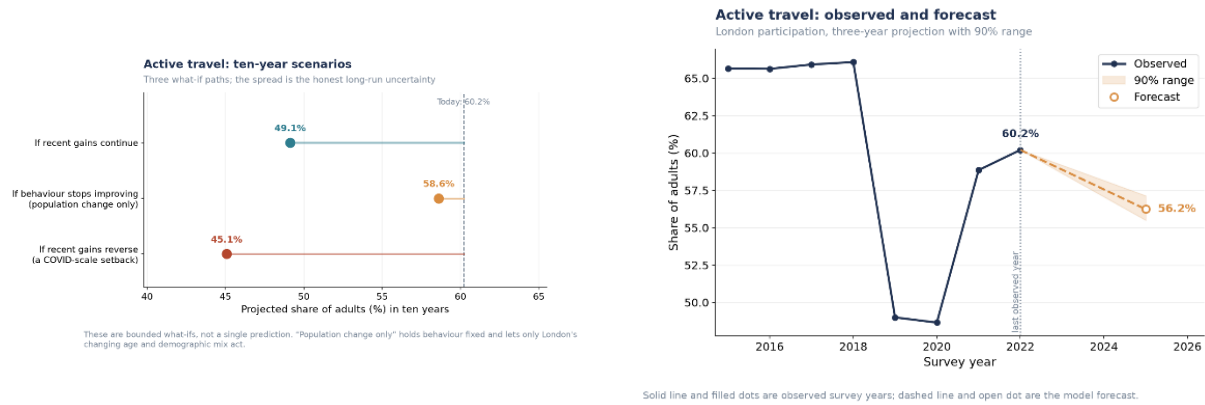


Figure 4.6: Walking: ten-year scenarios (largest-base activity measured).

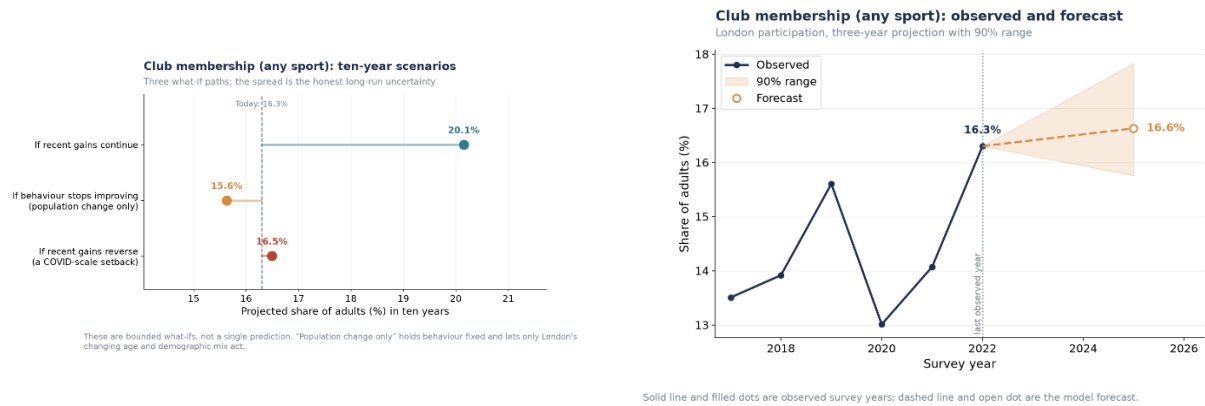


Figure 4.7: Club membership (any sport): ten-year scenarios.

Activity / Club Type	Today	If Gains Continue	Population Change Only	If Gains Reverse
Walking	75.7%	71.4%	74.6%	71.5%
Active travel	60.2%	49.1%	58.6%	45.1%
Fitness activities	35.4%	31.8%	34.5%	29.5%
Cycling	19.7%	20.1%	18.9%	17.2%
Team sports	8.2%	6.5%	7.7%	4.9%
Dance	8.9%	5.2%	8.7%	7.0%
Racket sports	6.1%	5.1%	6.0%	4.0%
Club membership (any sport)*	16.3%	20.1%	15.6%	16.5%
Fitness club / gym	10.0%	9.4%	9.5%	9.5%
Team-sport club	4.7%	7.1%	4.3%	4.7%

Table 4.3: Ten-year scenario range by activity/club type (% of London adults).

Note: * Club membership figures represent separate, potentially overlapping breakdowns rather than components of a single total. "Fitness club/gym" and "team-sport club" do not sum to "club membership (any sport)", since a person could belong to both, or to a club type not itemised here.

Continuing recent trends, as summarised in Table 4.3, suggests a decline in most individual activities, except cycling, while all three club-membership measures show potential growth. However, this recovery is uneven: team-sport club membership shows the strongest increase (4.7% → 7.1%), whereas fitness/gym membership remains broadly flat or declines (10.0% → 9.4%). The wide scenario ranges for team sports and active travel reflect their strong disruption during COVID, so these long-term projections should be

interpreted as directional rather than precise forecasts.

Forecasts of activity levels for each London borough. Applying the propensity model to each borough’s projected demographic composition produces a 2025 borough-level forecast ranging from 51.3% in Barking and Dagenham to 75.4% in Wandsworth.

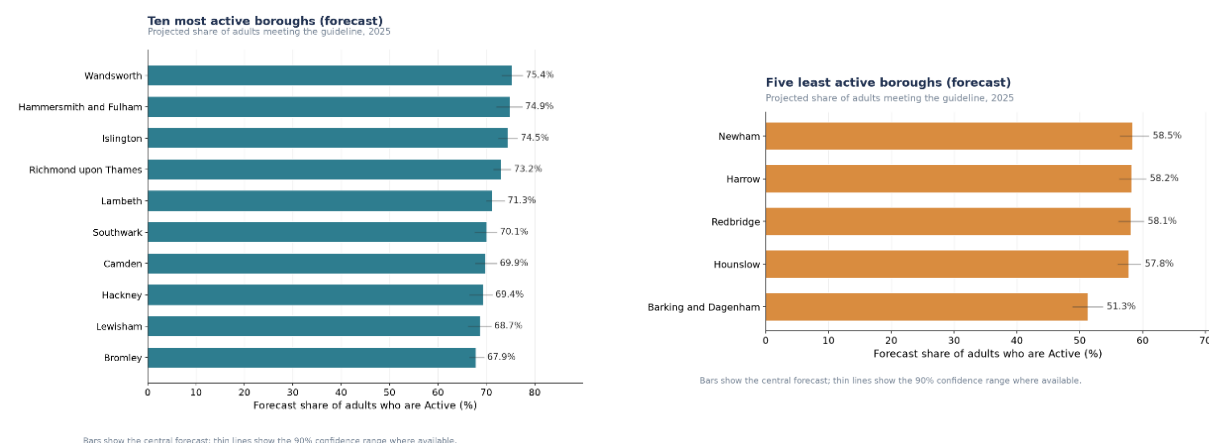


Figure 4.8: Ten most active boroughs and five least active boroughs, projected share of adults meeting the guideline, 2025.

The roughly 24-percentage-point gap between the most and least active boroughs is expected to persist throughout the projection period. The non-overlapping confidence intervals suggest that this difference is unlikely to be due to sampling variation alone. In addition, all 32 boroughs show a negative composition effect, ranging from close to zero to around -0.9 percentage points, indicating that projected demographic changes are expected to place some downward pressure on activity levels across London, although the size of this effect varies between boroughs. 4.8.

Key factors that may reduce free time and affect participation. Free-time constraint questions were asked once, in the 2022–23 wave only, of approximately a third of respondents. This section is therefore necessarily a cross-sectional snapshot rather than a ten-year forecast, as shown in Figures 4.9 and 4.10.

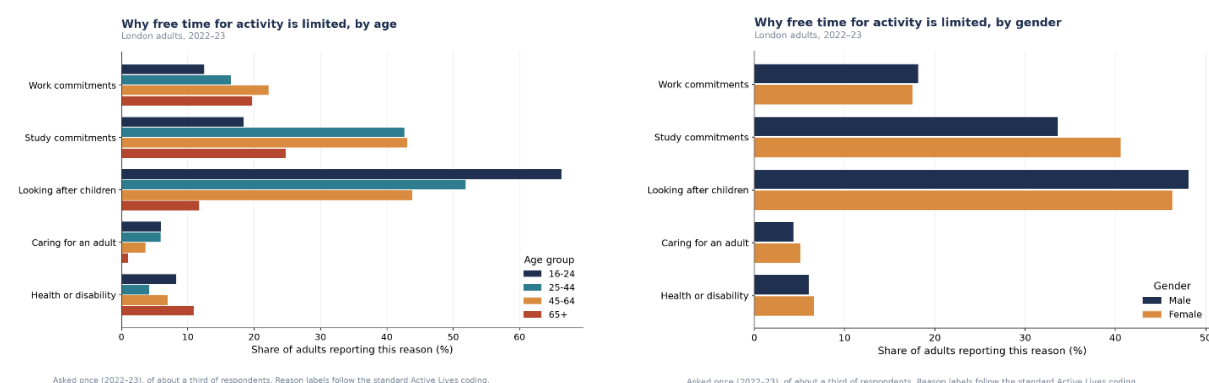


Figure 4.9: Reasons free time for activity is limited, by age group and gender (London adults, 2022–23).

The results show that work, study and childcare are the main reported constraints on free time, particularly among working-age adults, while caring responsibilities and health or disability are less frequently reported. However, reporting any free-time constraint was not associated with lower overall activity; constrained respondents were slightly more likely to be Active. This should be interpreted as a descriptive association rather than a causal relationship, as different constraints may have opposing effects. In particular, long working hours show a clearer negative association with physical activity, while activities such as childcare or active travel may themselves involve physical activity.

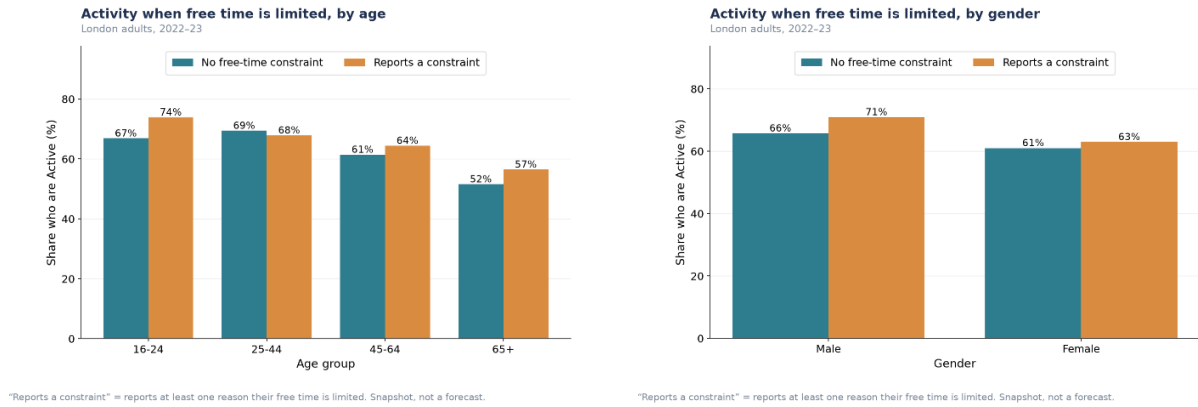


Figure 4.10: Activity rate by whether free time is constrained, by age group and gender.

4.3.2 P2 – Demographic-Specific Insights

Forecasted participation levels by demographic group. Figure 4.11 shows the forecast broken down by demographic group.

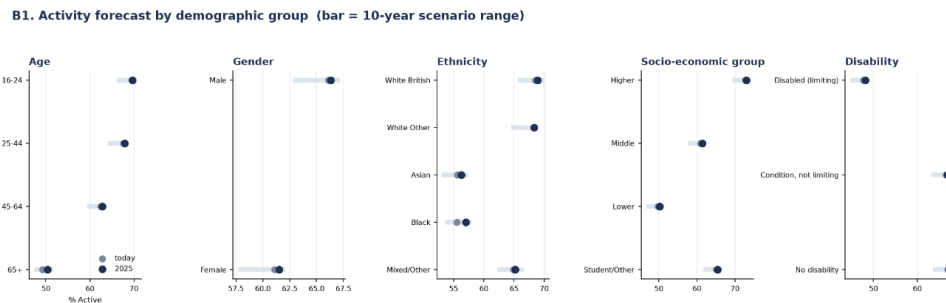
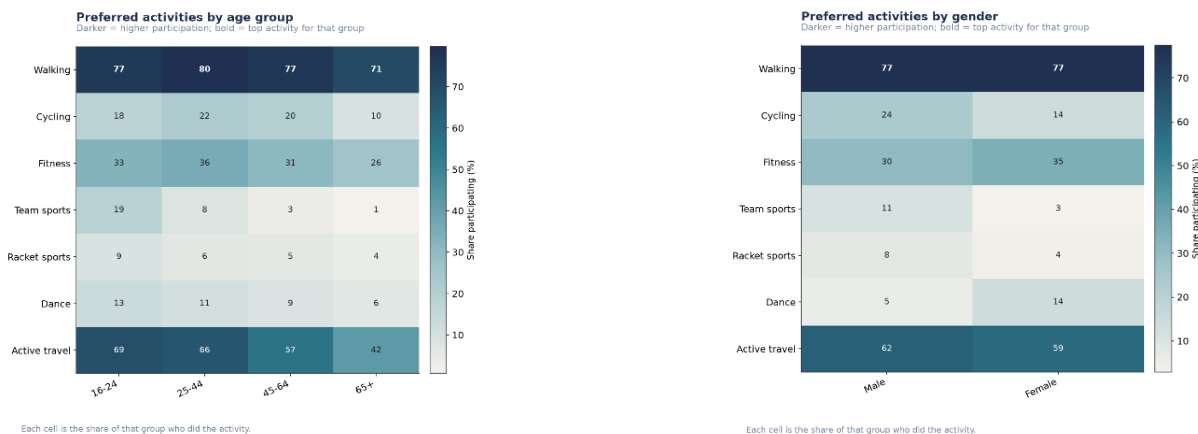


Figure 4.11: Activity forecast by demographic group; bar = 3-year forecast range, dot = today's rate.

The forecast shows clear differences in physical activity across demographic groups, with socio-economic position showing the widest gap: around 73% activity among the Higher group compared with 50% in the Lower group (23 percentage points). This is followed by the age gap of around 20 points (70% among 16–24 vs 50% among 65+), the disability gap of 19 points, and the ethnicity gap of 13 points. Gender shows the smallest difference, at around 4 points (66% male vs 62% female). The 65+ group is particularly important, as it is both the least active group and shows little projected change (around +1 percentage point), meaning London's ageing population may create a demographic headwind to overall activity levels.

Types of sport or physical activity each group is most likely to engage in. Figure 4.12 breaks down activity type by demographic group.



4.3. PHYSICAL ACTIVITY AND VOLUNTEERING TRENDS OVER THE NEXT TEN YEARS

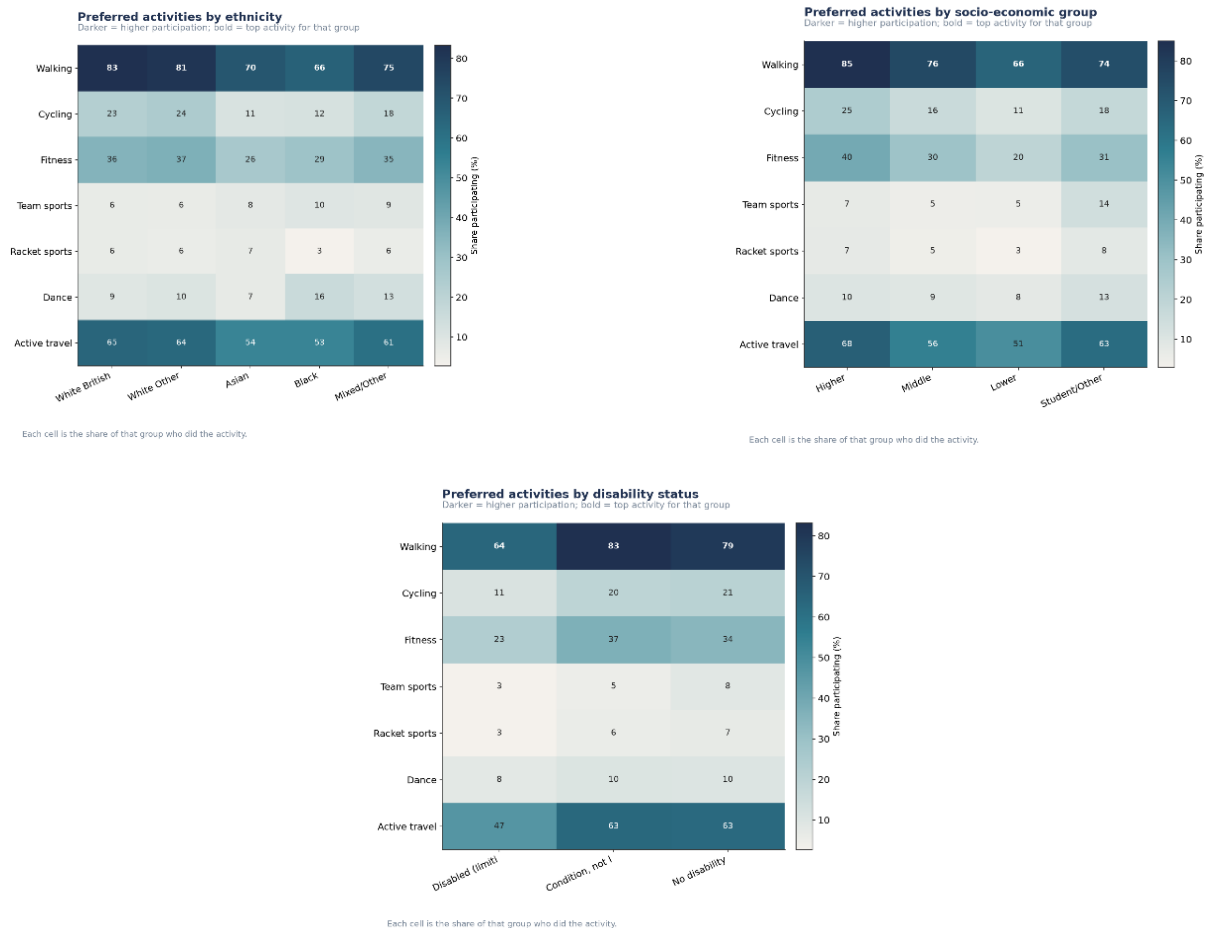


Figure 4.12: Which activities each group does most; bold marks the top activity for that group.

Walking is the most common activity across all demographic groups, although participation is lowest among adults with a limiting disability (64%). The strongest age gradient occurs in team sports, falling sharply from 19% among 16–24-year-olds to just 1% among those aged 65+, while active travel, the second most common activity, also declines from 69% to 42%. Gender differences vary by activity: men participate more in cycling, team and racket sports, while women lead in fitness and particularly dance (14% vs 5%), with walking equal at 77%. Ethnic differences are also activity-specific, with Black respondents showing the highest participation in both dance (16%) and team sports (10%), while Asian respondents have the lowest fitness (26%) and active travel (54%) rates.

Socio-economic position shows particularly large differences in facility-dependent activities, with fitness at 40% in the Higher group versus 20% in the Lower group and cycling at 25% versus 11%. However, team sports are highest among the Student/Other group (14%), while dance varies relatively little across socio-economic groups. Similarly, adults with a limiting disability have the lowest participation across all seven activities, with particularly large gaps in fitness, racket sports and active travel, but almost no gap in dance. Overall, dance stands out as the exception: unlike most other activities, it shows relatively small socio-economic and disability differences, suggesting that activities requiring greater access to facilities, equipment or organised provision are more strongly associated with demographic inequalities.

Participation within each London borough. **Data note:** the ask was for participation in specific activities within each borough. However, the details for activity-type $\times$ borough were not available at reportable cell sizes. The provided plot in Figure 4.13 is therefore broken down by age-group participation within each borough instead. Borough-by-demographic cells are small: a single borough-by-age cell contains around six respondents in a given wave. All eight waves were therefore pooled and thin cells suppressed.

The 65+ group is the least active in all 32 boroughs, but the youngest group is not always the most active. In 11 boroughs, the 25–44 group matches or exceeds the 16–24 group, most notably in **Borough 012 (Hackney)**, where activity rises from 64% (16–24) to 76% (25–44). The age gap also varies considerably

B3. Age gradient within each borough (all years pooled; dot = base too small)

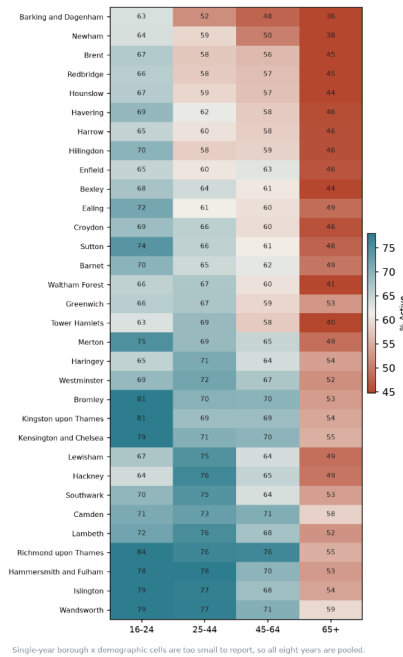
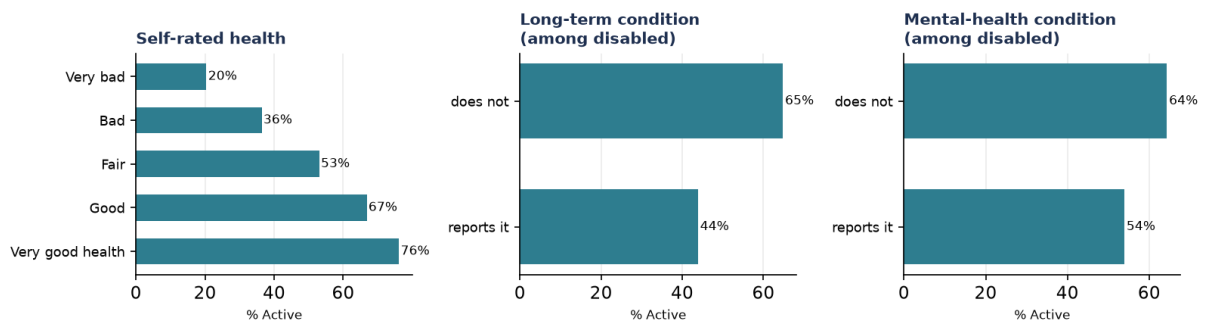


Figure 4.13: Age gradient within each borough, all waves pooled.

between boroughs, from 29 points in **Borough 027 (Richmond upon Thames)** (84% vs 55%) to just 11 points in **Borough 014 (Haringey)** (65% vs 54%). **Borough 032 (Wandsworth)** has the highest 65+ rate (59%), while **002 (Barking and Dagenham)** has the lowest (36%), showing that borough-level differences can reinforce the effect of age.

Likelihood of participation among people with health conditions. Self-rated health is available from 2018–19 onwards (only five years of data, with two years affected by COVID); the health-condition measures apply to people with a limiting disability. Results are descriptive, not forecasts, and are shown in Figure 4.14.

B4. Activity and health conditions (described, not forecast)



Self-rated health is collected from 2018-19 onward; long-term-condition flags apply to people with a limiting disability. Snapshot, not a projection.

Figure 4.14: Activity by self-rated health, long-term condition and mental-health condition.

Activity shows a strong and consistent health gradient, falling from 76% among those reporting very good health to 20% among those reporting very bad health. Among people with a limiting disability, those reporting a long-term condition are 21 percentage points less active than those without one (44% vs 65%), while the gap for mental-health conditions is smaller at 10 points (54% vs 64%). These results show a clear association between poorer health conditions and lower physical activity, although they do not establish causation.

4.3.3 P3 – Indoor vs Outdoor Activity

The expected balance of indoor versus outdoor physical activity. The projection extends only to 2025 and is based on a ~30% subsample, so it should be treated as indicative rather than a full ten-year forecast, as shown in Figure 4.15.

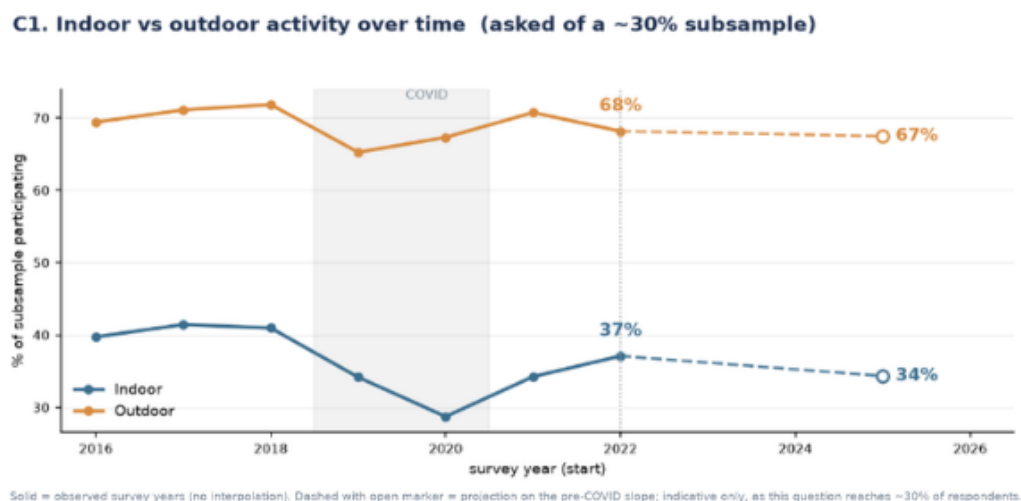


Figure 4.15: Indoor vs outdoor activity over time, observed 2016–2022 and projected to 2025.

Outdoor activity remains substantially more common than indoor activity, at around 65–72% compared with 29–42% over the observed period. Both declined during COVID, but indoor participation fell more sharply, from 41% in 2018 to 29% in 2020, and had only partly recovered to 37% by 2022. The indicative projection suggests indoor activity may fall further to 34% by 2025, while outdoor activity remains broadly stable at 67%, widening the gap between the two. However, this projection has a wide uncertainty range because it is based on the smaller subsample, so the figures should be interpreted directionally rather than as precise forecasts.

Differences in indoor and outdoor activity preferences among people with varying activity levels. Figure 4.16 shows how indoor and outdoor preference varies by activity level.

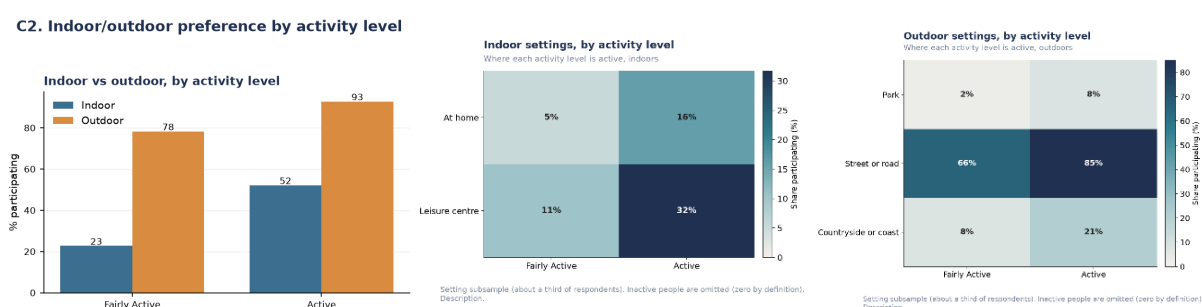


Figure 4.16: Indoor/outdoor preference by activity level, and where each activity level is active.

Indoor activity differs much more by activity level than outdoor activity: Active respondents are more than twice as likely to participate indoors (52% vs 23%), while outdoor participation is already high (93% vs 78%). Street/road activity dominates outdoors (66% vs 85%) and changes least, whereas leisure-centre use shows the largest absolute increase (11%→32%), followed by at-home activity (5%→16%). Park and countryside/coast use also increase (2%→8% and 8%→21%). Overall, Fairly Active people mainly rely on accessible outdoor activity, while Active people are more likely to add indoor and a wider range of activity settings to their routine.

Relationships between inner vs outer London locations, activity levels and socio-economic factors. Figure 4.17 presents the interaction between location, socio-economic group, and indoor/outdoor activity.

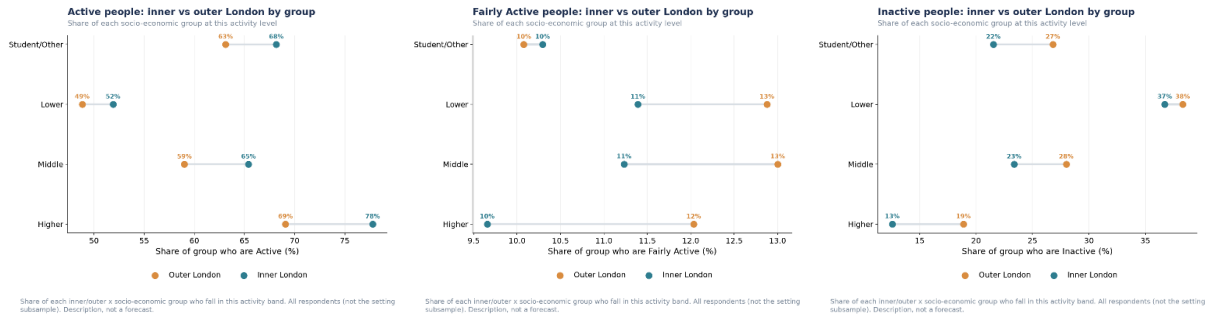


Figure 4.17: Inner vs outer London $\times$ socio-economic group, indoor and outdoor activity.

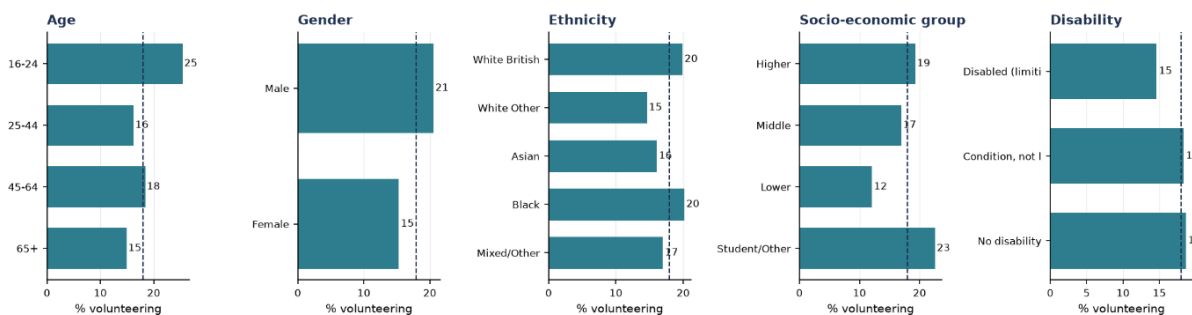
Inner London generally has a higher share of Active people than Outer London across all socio-economic groups, although the difference is relatively small for the Lower group (52% vs 49%) and larger for the Higher group (78% vs 69%). The reverse pattern is seen for inactivity, with Outer London consistently showing higher inactivity, particularly among the Higher group (19% vs 13%). Fairly Active rates are more mixed, with Outer London higher for the Lower and Middle groups but only slightly higher for the Higher group. Overall, socio-economic position remains the stronger divide, while location appears to widen or narrow these differences depending on activity level.

4.3.4 P4 – Volunteering in Sport

Data note: volunteering cannot be reliably forecast because the survey question base changed across the years and demographics explain only around 15% of borough-level variation. It is therefore reported descriptively rather than forecast at borough level.

Current volunteering patterns across demographic groups.

D1. Volunteering in sport by background (London average 18%; description, not forecast)



Pooled across waves. The volunteering question base changed during the series, so it is reported as a demographic profile, not a trend or a borough forecast.

Figure 4.18: Volunteering in sport by background, pooled across waves. Described, not forecast.

Volunteering is highest among younger adults, with 16–24-year-olds at 25%, compared with 15% among the 65+ group. Men also volunteer more than women (21% vs 15%), while there is a clear socio-economic gradient (19% Higher vs 12% Lower). In contrast, volunteering varies much less by ethnicity and disability, ranging from 15–20% across ethnic groups and 15–19% by disability status. Overall, age, gender and socio-economic background show stronger differences than ethnicity or disability.

Volunteer role patterns across demographic groups.

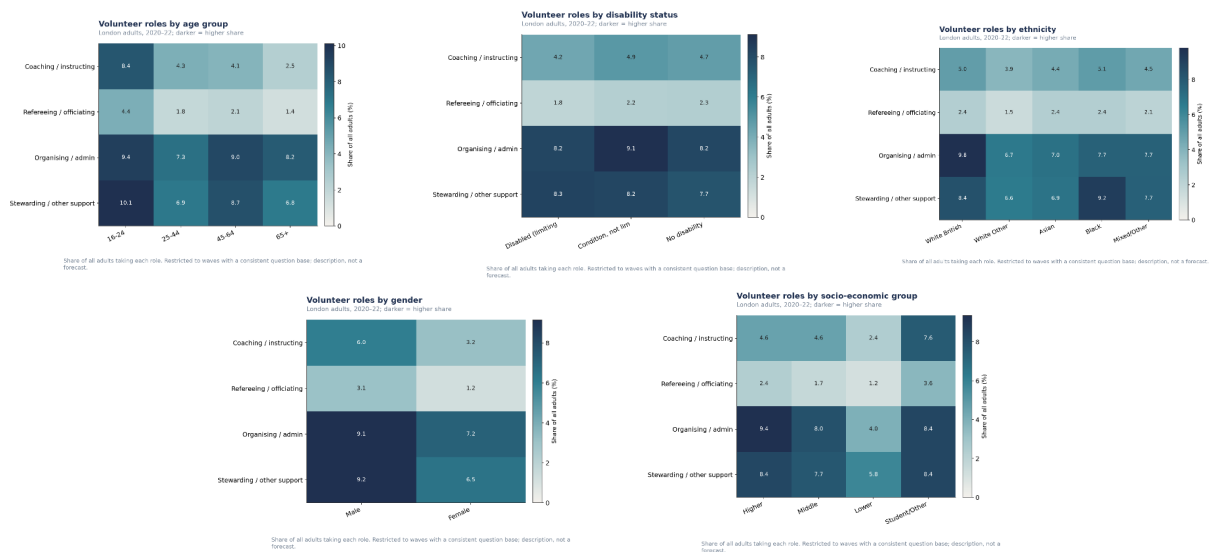


Figure 4.19: Volunteer roles by demographic group, waves 2020–22.

Organising/administration is the most consistent volunteer role across demographic groups, particularly among middle- and higher-income groups and most ethnicities. Younger adults stand out for more hands-on roles, with 16–24s highest in stewarding (10.1%) and coaching (8.4%). There is also a clear gender difference: men are more represented in coaching, officiating and stewarding, while women are more concentrated in organising and support. Lower socio-economic and some disability groups also show relatively higher reliance on stewarding/support roles.

4.4 Overall Synthesis

Taken together, the four parts of the analysis tell a fairly simple story about physical activity in London. London’s overall activity levels are not changing dramatically, but underneath this stable picture there are important differences between groups and places. An ageing population creates a challenge because older people are consistently the least active, while some younger and more active groups help balance this out.

Where people live also matters. The differences between boroughs become more noticeable for activities that need facilities, clubs or organised opportunities, such as indoor activity, leisure-centre use and sport. In contrast, simpler activities such as walking are more widely accessible. Volunteering shows the other side of this story: demographic differences are relatively small, suggesting that local opportunities and provision may matter more than who lives in an area.

Overall, the findings suggest that London Sport cannot rely on one solution. Demographic change needs to be planned for, while improving access to facilities, organised activities and local opportunities could provide more direct ways of encouraging participation.

4.5 Limitations

The findings should be interpreted with several limitations in mind. The survey is cross-sectional rather than longitudinal, so it captures population-level patterns rather than changes within the same individuals over time. COVID-19 also disrupted parts of the time series, reducing the number of suitable years for some forecasts.

Some outcomes, including volunteering and indoor/outdoor activity, have changes in question coverage or smaller subsamples and were therefore reported descriptively rather than as full ten-year forecasts. The analysis also identifies borough differences but cannot fully explain them without external data on factors such as facility availability, green space and cycling infrastructure. Finally, demographic forecasts should be treated as indicative estimates rather than precise predictions where those groups were not directly included in the population projections.

Chapter 5

Conclusion

5.1 Summary of Contributions

This project addressed a gap in how the Active Lives Survey has typically been used: existing research describes historical participation but rarely forecasts future change. Using eight harmonised waves of the Active Lives Adult Survey (2015–16 to 2022–23) across London’s 32 boroughs, the project developed a decision support framework that separates the relationship between individual demographic characteristics and activity level from projected changes in future population composition, avoiding the need to fit a conventional time series model to only 256 borough-year observations.

Key contributions include:

1. Harmonisation of eight structurally inconsistent survey waves (292–601 variables each) into a single dataset of 135,497 respondents with standardised ONS borough codes.
2. A propensity-based forecasting framework that decomposes forecasted change into a *composition effect* (demographic shift) and a *behaviour effect* (change within demographic groups), the project’s most policy-relevant finding, showing that London’s apparently stable overall activity rate masks an ageing-population headwind being offset by genuine behavioural improvement.
3. A four-model comparison (ordinal logistic regression, LightGBM, CatBoost with SHAP, K-Modes clustering, and Bayesian hierarchical modelling) under consistent time-based validation, with LightGBM selected as the final forecasting model (AUC 0.707, ~16 minute runtime).
4. A composition–place decomposition showing borough identity explains a further 40 percentage points of between-borough variation beyond demographics alone (48% → 88%).
5. Application of the framework across four pillars — overall participation, demographic differences, indoor/outdoor balance, and volunteering — forecasting only where the data supported it and reporting descriptively elsewhere.

5.2 Evaluation Against Aims and Objectives

All five objectives set out in Chapter 1 were met. Harmonisation (Objective 1) is documented in Chapter 3. The propensity model was developed and benchmarked (Objective 2): the ordinal logistic regression offered near-identical performance to LightGBM (AUC 0.704 vs 0.707) with fully transparent coefficients, but the fuller comparison in Chapter 4 — weighing runtime, non-linear interactions, and integration with forecasting — favoured LightGBM as the production model, a reasonable trade-off given it retains feature-importance interpretability while handling the broader activity-type dataset better.

Composition versus place effects (Objective 3) were established via the composition–place decomposition and corroborated independently by the Bayesian random-effects model and the CatBoost/SHAP residual analysis. The forecasting framework itself (Objective 4) is the project’s central methodological contribution, directly addressing the eight-wave data limitation that ruled out conventional time series methods. Finally, the framework was applied across all four analytical pillars (Objective 5), with volunteering and indoor/outdoor activity correctly handled descriptively rather than forecast, given their more limited question coverage.

The project therefore achieved its aim: an interpretable, borough-level forecasting framework was built and applied, producing results, notably the composition/behaviour decomposition and the persistent ~ 24 point gap between the most and least active boroughs, directly usable by a non-technical stakeholder such as London Sport.

5.3 Limitations and Future Work

The survey’s repeated cross-sectional design means the analysis captures population-level rather than within-individual change, and COVID-19 disruption reduced usable years for several sub-analyses (health-linked participation, indoor/outdoor split). Borough-level differences are not fully explained by demographic and survey data alone; incorporating external data such as facility density, green space, and cycling infrastructure is a natural next step to help distinguish environmental access from persistent local behaviour differences.

Two further directions stand out. First, the Bayesian hierarchical model showed promise for quantifying borough-level forecast uncertainty but was constrained by long runtimes (~ 96 minutes) and limited posterior sampling; a more efficient implementation could move it from a corroborating role to a primary source of uncertainty estimates. Second, as further survey waves become available, volunteering and indoor/outdoor measures such as currently descriptive due to changing question bases and smaller subsamples, could be brought into the forecasting framework once a longer, more consistent time series exists.

Bibliography

Appendix A

An Example Appendix

Content which is not central to, but may enhance the dissertation can be included in one or more appendices; examples include, but are not limited to

- lengthy mathematical proofs, numerical or graphical results which are summarised in the main body,
- sample or example calculations, and
- results of user studies or questionnaires.

Note that in line with most research conferences, the examiners are not obliged to read such appendices.